Department of Mathematics

MODULE II

Inference for Decision Making – I

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1. Introduction to hypothesis

A statistical hypothesis is a statement (an assumption) about the probability distribution of a random variable (population).

Suppose there is a brand of drug for a certain disease and from the experience, we know that the proportion of patients who get cured using this drug is $p_0 = 0.5$. Now, a new drug is invented and it is to be tried. It is of interest to the drug control authorities to know if the effectiveness of the new drug is more than the existing one. Let p denote the proportion of the patients who will get cured using the new drug then they are interested to check if $p > 0.5$. This is a problem of testing of hypothesis. Suppose we have measurements on weights of new born kids in an ethnic group. We want to test whether the measurement follow a normal distribution, this is a statistical hypothesis.

1.1 Characteristics of Hypothesis

For hypotheses to be worthwhile and testable, they must fulfill some important conditions. These conditions refer to as characteristics of hypotheses include:

(i) Clarity: Hypotheses must be clear. The researcher must define all the variables used in the hypotheses conceptually and operationally.
(ii) Specification or Direction: Hypotheses must be specific. The researcher must point out the expected relations among the variables as to ether they are positive or negative and the conditions under which the relations will hold.
(iii) Measurability: Hypotheses must be testable with available methods. This means that the researcher must ensure that there are available methods or techniques to test or evaluate his hypotheses.

(iv) Testability: Hypotheses must be value free. Because research in the social sciences takes place in the social milieu, the researcher must be aware of personal biases and make them as explicit as possible

(v) Terrain: Every hypothesis must be drawn to cover a known population or a sample of it. The terrain must be indicated to show the area of application. Some hypotheses may apply to a population and may not be applicable to another population even though these populations have some similar characteristics.

1.2 Function of Hypothesis

On a broad basis, hypotheses play the role of a torchlight that illuminates and directs the researcher towards arriving at the exact causal factors to identified problems. The following are the functions of hypotheses;

(i) Help to build a theory by confirming or rejecting appropriate aspects of a given theory.
(ii) They help the researcher to test a theory for validity and reliability.
(iii) Hypotheses assist the researcher in the gathering of relevant data within the limits of the research problem
(iv) Hypotheses help to facilitate the differentiation between scientific theories and sets of generalizations through testing.
(v) Hypotheses facilitate the drawing of scientific and needed conclusions from scientific tests.
(vi) They help to provide direction to a scientific research or inquiry.

1.3 Types of Hypothesis

The hypothesis mainly two types that are

1. Simple hypothesis
2. Composite hypothesis

If a hypothesis completely specifies a given probability distribution, the it is called a simple hypothesis, otherwise it is called a composite hypothesis. Suppose random variables $X_1, X_2, \dots, X_n$ are follow the normal distribution with mean μ and standard deviation σ (N (μ, σ^2)).

Example of composite hypothesis is

$$\begin{cases} H_0: \mu = 0 \\ H_1: \mu = 1 \end{cases} \quad (\sigma \text{ is unknown})$$

and simple hypothesis is

$$\begin{cases} H_0: \mu = 0, \ \ \sigma = 1 \\ H_1: \mu = 1, \ \ \sigma = 1 \end{cases} \quad (\mu \text{ and } \sigma \text{ both are known})$$

1.4 Null and Alternative Hypothesis

The null hypothesis is a hypothesis which is tested for possible rejection under the assumption that it is true. If the hypothesis is not null then this hypothesis is called alternative hypothesis.

Suppose in a coin tossing experiment, we want to test whether a coin is unbiased or not. Let p denote the probability of occurrence of a head. Then, we want to test if p= $\frac{1}{2}$ or p≠ $\frac{1}{2}$.

Now, we denoted the null hypothesis as H_0 and Alternative hypothesis H_1. So, we can write as for the above experiments

$H_0\ (null)$: p= $\frac{1}{2}$

$H_1\ (Alternative)$: p≠ $\frac{1}{2}$.

1.5 Critical or Acceptance Regions

Let S be the sample space of the random experiment. A critical region or the rejection region of the test is that past of the sample space that corresponding to the rejection of the null hypothesis.

If X∈ S_R, we reject H_0, then S_R is called the rejection region or critical region. Other part, if X ∈ $S_R^c = S_A$, we do not reject H_0, then S_A is called Acceptance region.

Example: We have a coin. We toss it thrice to test whether the probability of head is $\frac{1}{4}$ or $\frac{3}{4}$. Then,

The sample space is S = {HHH, HHT, HTH, THH, HTT, THT, TTH, TTT}.

$$\begin{cases} H_0: \text{p} = \frac{1}{4} \\ H_1: \text{p} = \frac{3}{4} \end{cases}$$

Now, $S_A = \{HTT, THT, TTH, TTT\}$ and $S_R = \{HHH, HHT, HTH, THH\}$.

i.e., if 2 or 3 heads are observed we go in favour of H_1 and if 0 or 1 head is observed we go in favour of H_0.

1.6 Types of Error

When we conduct a test of hypothesis, we are likely to make two types of errors:

Type-I Error: Rejecting H_0 when it is true. (First Kind)

Type-II Error: Accepting H_0 when it is false. (Second Kind)

	H_0 is true	**H_0 is false**
Do not reject H_0	Correct decision	Type II error
reject H_0	Type I error	Correct decision

1.7 Level of Significance

The significance level (α) is the threshold for rejecting the null hypothesis. It represents the maximum allowable probability of Type I error, the risk of incorrectly rejecting a true null hypothesis. The Type II error rate (β) is the probability of failing to reject the null hypothesis when it is actually false. i.e.,

$$\alpha = P(\text{Type I error}) = P(\text{Reject } H_0, \text{ when it is true}) = P(X \in S_R)$$

$$\beta = P(\text{Type II error}) = P(\text{Accepted } H_0, \text{ when it is false}) = P(X \in S_A)$$

In other words, it represents the likelihood of not detecting a true effect of difference when one exists. It is complementary to the statistical power of a test, which is the probability of correctly rejecting the null hypothesis when it is false. The level of significance usually employed in testing of hypothesis are 5% and 1%. It is always fixed in advance before collecting the sample information.

1.8 Central Limit Theorem

Let $X_1, X_2, \ldots, X_n$ be a random sample of size n drawn independently from a population having mean μ and finite variance σ^2. Then, for a sufficiently large sample size n,

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0,1).$$

1.9 One-tailed and Two-tailed tests

In any test, the critical region is represented by a portion of the area under the probability curve of the sampling distribution of the test statistics.

1.9.1 One-tailed Test

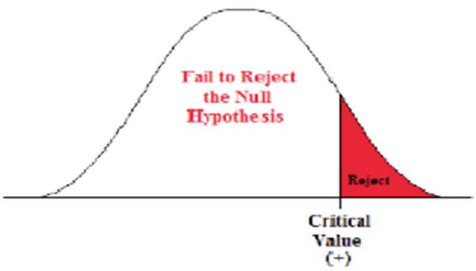
A test of any statistical hypothesis where the alternatives hypothesis is one tailed (left or right tailed) is called a one-tailed test. For example, a test for testing the mean of a population $H_0: \mu = \mu_0$ against the alternative hypothesis:

$H_1: \mu > \mu_0$ (Right-tailed) or $H_1: \mu < \mu_0$ (Left-tailed)

On the right-tailed test, the critical region lies entirely in the right tail of the sampling distribution of $\bar{x}$, while for the left-tailed test, the critical region is entirely in the left tail of the distribution.



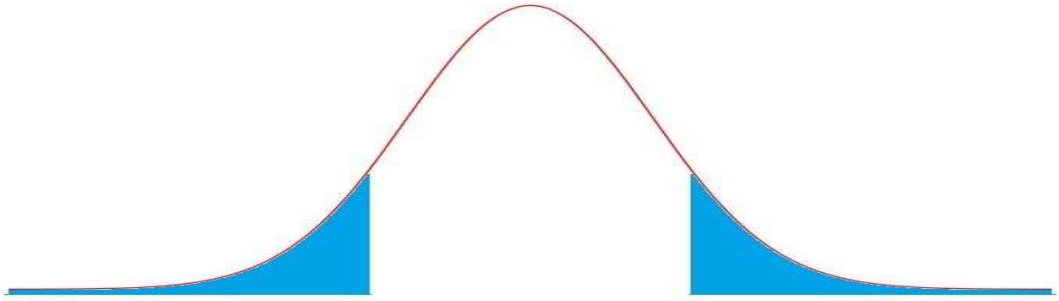
Left-tailed test



Right-tailed test

1.9.2 Two-tailed Test

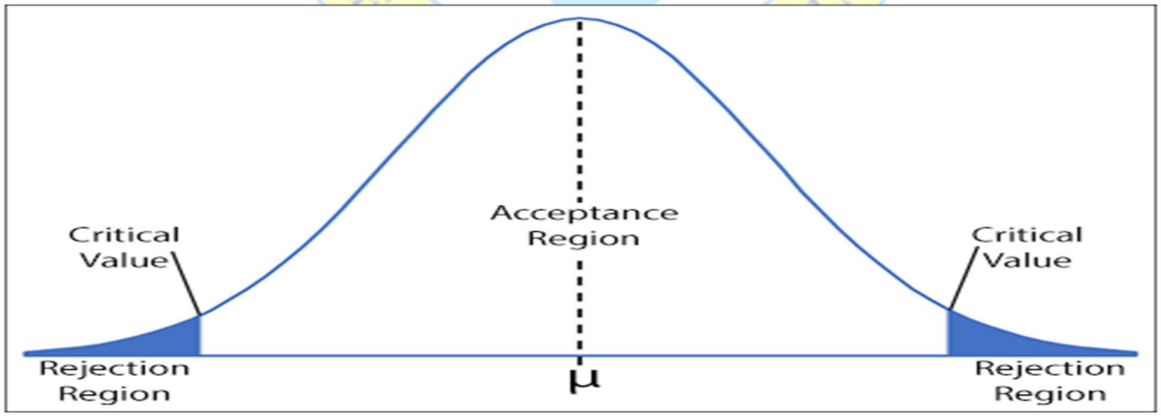
A test of statistical hypothesis where the alternative hypothesis is two-tailed such as $H_0: \mu \neq \mu_0$, against the alternative hypothesis $H_1: \mu \neq \mu_0$ ($\mu > \mu_0$ and $\mu < \mu_0$), is known as two-tailed test and in such a case the critical region is given by the portion of the area lying in both tails of the probability curve of the test statistic.



1.10 Critical Values or Significant Values

The value of test statistic which separates the critical (or rejection) region and the acceptance region is called the critical value or significant value. It depends upon:

(i) The level of significance used, and
(ii) The alternative hypothesis, whether it is two-tailed or single-tail



For the large sample (n>30), the value of Z is given for the statistic t as

$$Z = \frac{t - E(t)}{s \times E(t)} \sim N(0,1)$$

The value of Z under the null hypothesis is known as test statistic. The critical value of the test statistic at level of significance α for a two-tailed test is given by Z_α is determined by the equation:

$P(|Z| > z_\alpha) = \alpha \Rightarrow P(Z > z_\alpha) + P(Z < -z_\alpha) = \alpha \Rightarrow 2P(Z > z_\alpha) = \alpha \Rightarrow P(Z > z_\alpha) = \frac{\alpha}{2}$.

Thus, the area of each fail is $\frac{\alpha}{2}$.

For single-tail alternative,

$P(Z > Z_\alpha) = \alpha$ for right-tailed test

$P(Z < -Z_\alpha) = \alpha$ for left-tailed test.

Hence, the significant or critical value of Z for a single-tailed test at level of significance "α" is same as the critical value of Z for a two-tailed test at level of significance "2α".

1.11 Tests of Significance for large Samples

For large sample values, almost all the distributions are very closely approximated by normal distribution. Thus, in this case the normal test will be applied, which is based upon the following fundamental property of the normal probability curve. If X~ N(μ, σ), then

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1).$$

Thus, from the normal probability tables, we have

$P(|Z| \leq 3) = P(-3 \leq Z \leq 3) = 0.9973$

$\Rightarrow P(|Z| > 3) = 1 - P(|Z| \leq 3) = 1 - 0.9973 = 0.0027$.

i.e., in all probability, we should expect a standard normal variate to lie between ± 3.

Also, from the normal probability tables, we have

$P(|Z| \leq 1.96) = P(-1.96 \leq Z \leq 1.96) = 0.95$

$\Rightarrow P(|Z| > 1.96) = 1 - P(|Z| \leq 1.96) = 1 - 0.95 = 0.005$.

And

$P(|Z| \leq 2.58) = 0.99$

$\Rightarrow P(|Z| > 2.58) = 0.01$.

For a two-tailed test, if |Z|>1.96, then H_0 is rejected at 5% level of significance. Again if, |Z|>2.58, then is rejected at 1% level of significance and if |Z|≤ 2.58, H_0 may be accepted at 1% level of significance.

From the normal probability tables, we have

P(Z>1.645)=0.5- P(0≤Z≤ 1.645) $= 0.5 - 0.45 = 0.05$ $(or\ 5\%)$

P(Z>2.33)=0.5- P(0≤Z≤ 2.33) $= 0.5 - 0.49 = 0.01$ $(or\ 1\%)$.

Hence, for a single-tailed test, we compare the computed value of |Z| with 1.65 (at 5% level of significance) and 2.33 (at 1% level of significance) and accept or reject H_0 accordingly.

Notes:

i. If |Z|>3, the H_0 is always rejected.
ii. Comparison Table Below.

Critical Value (z_α)	Level of Significance		
	1%	5%	10%
Two-tailed test	$\|z_\alpha\| = 2.58$	$\|z_\alpha\| = 1.96$	$\|z_\alpha\| = 1.645$
Right-tailed test	$\|z_\alpha\| = 2.33$	$\|z_\alpha\| = 1.645$	$\|z_\alpha\| = 1.28$
Left-tailed test	$\|z_\alpha\| = -2.33$	$\|z_\alpha\| = -1.645$	$\|z\| = -1.28$

2. Hypothesis Testing Procedure

There are two types of statistical inferences namely: estimation of population parameters and hypothesis testing. Hypothesis testing is one of the most important tools of the application of statistics to real life problems. Most often, decisions are required to be made concerning populations on the basis of sample information. Statistical tests are used in arriving at these decisions. A statistical hypothesis test is a method of making statistical decisions using experimental data. We use hypothesis testing to decide whether experimental results contain enough information to cast doubt on conventional wisdom.

Often in social research methodology, we talk about hypothesis testing and this process is applicable in all of the social sciences no matter the discipline. In fact, the application of scientific methods of inquiry into such hitherto non-numeric disciplines like political science, sociology and mass communication has further made these disciplines to become much scarier to students who have little background of mathematics and particularly statistics. Since the common variable in all these disciplines is human behavior, scientific inquiry in these disciplines often involves the measurement of some relationships relating to human behavior. Consider the following pairs of relationships:

a. Voting pattern & ethnicity
b. Workers' punctuality & salary increase
c. Police brutality & regime type
d. Voice expression & audience applause

e. Price increase & the demand for Substitutes
f. Climatic change & human settlement

You will observe that in each of the above relationships, one variable is *dependent* (that is, changes in it can only occur when the other variable has changed) while the other is *independent* (that is, one that causes changes to the other variable). The dependent variable in each relationship context is known as the *explained* or *endogenous* variable while the independent variable in such relationship contexts is the *explainer* or *exogenous* variable.

There are five steps to any statistical test as outlined below.

Steps in Hypothesis Testing

1. Identify the null hypothesis H_0 and the alternate hypothesis H_1.
2. Choose a critical (rejection) region or the p-Value level of Significance (α)
3. Select the test statistic and determine its value from the sample data.
4. Compare the observed value of the statistic to the critical value obtained for the chosen α.
5. Make a decision.

1. **Identify the null hypothesis H_0 and the alternate hypothesis H_1.**

In attempting to reach a decision, it is useful to make an educated guess or assumption about the population involved, such as the type of distribution. To test the relationship between any given pair of variables, we must first state two sets of hypotheses connecting these variables: one a null hypothesis and the other an alternate hypothesis to be tested. It is important to correctly state these hypotheses. Let us consider the first set of variables given above and derive a null and an alternate hypothesis as follows:

Null Hypothesis:

H_0: There is no significant relationship between the ethnic affiliation of voters in a given constituency and their voting pattern.

Alternative Hypothesis:

H_1: There is a significant relationship between the ethnic affiliation of voters in a given constituency and their voting pattern

In the above statements, we can observe that the two variables (one dependent and the other independent) have been connected together.

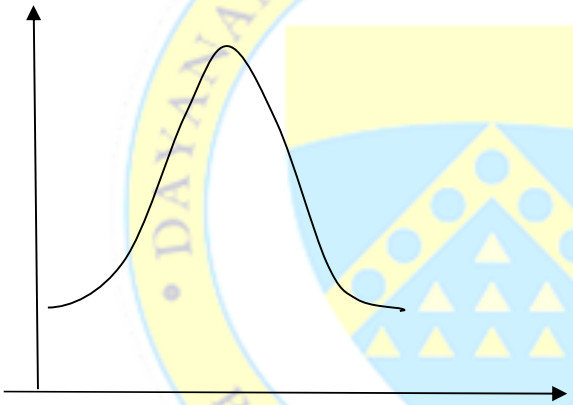
2. Choose a critical (rejection) region or a p-value level of Significance (α)

The critical region is the part of the sample space where the null hypothesis H_0 is rejected. The size of this region is determined by the probability (α) of the sample point falling in the critical region when H_0 is true. This region is also known as the level of significance, the probability of the value of the random variable falling in the critical region. The value should be small, usually less than 10%. It is important to consider the consequences of both types of errors. It is on the basis of this level of significance that "Statistical significance" derives. Statistical significance refers only to the rejection

of a null hypothesis at some level (α). It implies only that the observed difference between the sample statistic and the mean of the sampling distribution did not occur by chance alone.

3. Select the test statistic and determine its value from the sample data.

The test statistic is the random variable X whose value is tested to arrive at a decision. This value is called the observed value of the test statistic. Depending on the type of test the inquirer is conducting, several test statistics are available to test a null hypothesis for a given set of data (distribution). In doing tests of association, the Chi Square statistic, the Correlation coefficient (*r*) or the *T-statistic* (*t*) can be used if the number of samples in the distribution is not more than 30. Often, a *T- statistic* is usually appropriate for a small number of samples; but for larger number of samples (that is, $n \geq 30$), a *Z-statistic* can be used if data are normally distributed since quite often large samples are not easy to obtain, in which case the *t*-distribution can be used. The Central Limit Theorem states that for large sample sizes ($n \geq 30$) drawn randomly from a population, the distribution of the means of those samples will approximate normality, even when the data in the parent population are not distributed normally. The population standard deviation δ is estimated by the sample standard deviation, *s*. The *t*-curves are bell shaped and distributed around t=0 as shown in the diagram below.



The exact shape on a given t-curve depends on the degrees of freedom. In a case in which test requires performing multiple comparisons, we can use the Analysis of Variance techniques (ANOVA) by one way or two way in which case the *F*-statistic is normally used. The F-statistic is defined as the *ratio of the mean square due to the variability between groups to the mean square due to the variability within groups*.

4. Compare the observed value of the statistic to the critical value obtained for the chosen critical region (α)

The critical values of each of these statistic stated above in this section is then read off from standard statistical tables on the appropriate distribution knowing the Type-I error and the degrees of freedom. The degrees of freedom (df), is equal to the sample size (n) minus one. Thus, $df = n - 1$. When

comparing observed (calculated) values of test statistic with the critical (standard table) values of that statistic, we do so at the relevant degree of freedom. Usually, at the point of comparison, we are about to make either of two decisions namely to reject a null hypothesis H_0 in favour of the alternate hypothesis H_1 or to accept a null hypothesis H_0 in favour of the alternate hypothesis H_1. Often the result of the comparison is presented in a particular fashion.

5. Make a Decision and Interpret your Results

Decide to either **fail to reject** the null hypothesis or **reject** it in favor of the alternative. The decision rule is to reject the null hypothesis H_0 if the observed value T_{obs} or X_{obs} or F_{obs} or r_{obs} or t_{obs} as the appropriate case may be is in the critical region, and to accept or "fail to reject" the hypothesis if otherwise. If the test statistic falls in the rejection/critical region, H_0 is rejected, else H_0 is accepted. The acceptance or rejection of the null hypothesis helps you to make appropriate suggestions for policy making if such variables are available for control by the appropriate authorities.

3. The p-value Approach to Hypothesis Testing

New Approach an alternative way to conclude a test of hypotheses is to compare the p-value of the sample test statistic with significance level. The p-value of the sample test statistic is the smallest level of significance for which we can reject H_0. The p-value is compared to significance level α, and on this basis H_0 is either rejected or not rejected. Therefore, if p-value $\leq \alpha$, we reject H_0, and if p-value $> \alpha$, we do not reject H_0. The advantage of knowing the p-value is that we know all level of significance for which the observed sample statistic tells us to reject H_0. To alleviate the problems encountered in the classical approach to hypothesis testing.

Interpretation of p-value

The computation of a p-value is probably the most common approach for summarizing the results of a statistical test. The p-value, which directly depends on a given sample, attempts to provide a measure of the strength of the result of a test, in contrast to a simple reject or do not reject. If the null hypothesis is true and the chance of random variation is the only reason for sample differences, then the p-value is a quantitative measure to feed into the decision making process as evidence. The p-value should be interpreted carefully. Although, the p-value has straightforward interpretation it is also mis-understood quantity in statistics. Four usual mis-interpretation of the p-values must be emphasized; i.e. p-value is not.

- The probability that null hypothesis is true.
- The probability of committing type I error.
- The probability of making wrong decision
- The probability that the sample statistic is due to chance alone.

Information about the significance level is usually given in the form of the p-value inequality for the most informative a level. For example, if $p < 0.01$, then p is necessarily also less than 0.05 ($p < 0.05$). On the other hand, knowing that $p < 0.05$ does not indicate whether p is also less than 0.01or not; hence, $p < 0.01$ is more informative than $p < 0.05$ if both are true. Similarly, $p > 0.05$ is more informative than $p > 0.10$ if both are true.

Solved Problem 1: A manufacturer claims that a new process improves the mean weight of their product packages from the previous mean of 100 grams. A random sample of 49 packages is taken, and the sample mean weight is found to be 103 grams. Assume that the population standard deviation is 10 grams. Test whether the new process has improved the mean weight of the packages at the 5% significance level.

Solution:

Step 1: State the Hypotheses

The manufacturer claims that the new process has increased the mean weight. Therefore, the alternative hypothesis is that the population mean is greater than 100 grams.

The hypotheses are:

$$H_0: \mu = 100$$
$$H_1: \mu > 100$$

where

- H_0: The new process does not improve the mean weight.
- H_1: The new process improves the mean weight.

Since the alternative hypothesis uses the symbol ">", this is a right-tailed test.

Step 2: Choose the Level of Significance

The level of significance is given as

$$\alpha = 0.05.$$

This means we are willing to accept a 5% chance of rejecting the null hypothesis when it is actually true.

Step 3: Compute the Test Statistic

Since the population standard deviation is known, the Z-test statistic is

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}.$$

Substitute the given values:

$$z = \frac{103 - 100}{10/\sqrt{49}}$$

we obtain

$$z = \frac{3}{10/7} = \frac{3}{1.4286} = 2.10.$$

Thus, the calculated test statistic is

$$\boxed{z = 2.10.}$$

Step 4: Determine the Critical Value

For a **right-tailed Z-test** at the **5% significance level**, the critical value from the standard normal distribution table is

$$z_{0.05} = 1.645.$$

This means:

- Reject H_0 if $Z > 1.645$.
- Otherwise, do not reject H_0.

Step 5: Make the Decision

The calculated test statistic is

$$z = 2.10,$$

and the critical value is

$$1.645.$$

Since

$$2.10 > 1.645,$$

the test statistic falls in the rejection region.

Therefore, we **reject the null hypothesis**.

Alternatively, the p-value from table is

$$P(Z > 2.10) \approx 0.0179.$$

Since

$$0.0179 < 0.05,$$

we again reject the null hypothesis.

Step 6: State the Conclusion

At the 5% level of significance, there is sufficient statistical evidence to conclude that the new manufacturing process has significantly improved the mean weight of the product packages.

- **Conclusion:**

There is sufficient statistical evidence at the **5% significance level** to conclude that the **new process has increased the mean weight of the product packages**. Therefore, the manufacturer's claim is accepted.

Solved Problem 2: A drug company is testing a drug intended to increase heart rate. A sample of 100 yielded a mean increase of 1.4 beats per minute, with a standard deviation known to be 3.6. Since the company wants to avoid marketing an ineffective drug, it proposes a 0.01 significance level. Should it market the drug?

Solution: Since the population standard deviation is known, we use a **one-sample Z-test** for the population mean.

Step 1: State the Hypotheses

The company wants to determine whether the drug **increases** the heart rate.

$$H_0: \mu = 0$$
$$H_1: \mu > 0$$

where μ is the true mean increase in heart rate.

Step 2: Level of Significance

$$\alpha = 0.01$$

This is a **right-tailed test**.

Step 3: Test Statistic

Use the Z-test statistic:

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

Substituting the given values,

$$z = \frac{1.4 - 0}{3.6/\sqrt{100}} = \frac{1.4}{3.6/10} = \frac{1.4}{0.36} = 3.89.$$

Step 4: Critical Value

For a right-tailed test at the 1% significance level,

$$z_{0.01} = 2.326.$$

Step 5: Decision Rule

- Reject H_0 if $Z > 2.326$.
- Otherwise, do not reject H_0.

Since

$$3.89 > 2.326,$$

we **reject the null hypothesis**.

Step 6: Conclusion

There is sufficient evidence at the **1% significance level** to conclude that the drug **increases the heart rate**.

Or

p-value Approach

Step 1: State the Hypotheses

The company wants to determine whether the drug increases the heart rate.

$$H_0: \mu = 0$$
$$H_1: \mu > 0$$

This is a right-tailed Z-test.

Step 2: Compute the Test Statistic

The test statistic is

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}.$$

Substituting the given values,

$$z = \frac{1.4 - 0}{3.6/\sqrt{100}} = \frac{1.4}{3.6/10} = \frac{1.4}{0.36} = 3.89.$$

Step 3: Calculate the p-value

The p-value for a right-tailed test is

$$\text{p-value} = P(Z > 3.89).$$

From the standard normal distribution table,

$$P(Z > 3.89) = 1 - P(Z < 3.89).$$

Since

$$P(Z < 3.89) = 0.99995,$$

we obtain

$$\boxed{\text{p-value} = 1 - 0.99995 = 0.00005.}$$

Step 4: Decision Rule

Compare the p-value with the significance level.

- p-value $= 0.00005$
- $\alpha = 0.01$

Since

$$0.00005 < 0.01,$$

Reject the null hypothesis H_0.

Step 5: Conclusion

There is sufficient statistical evidence at the 1% significance level to conclude that the drug increases the mean heart rate.

4. Hypothesis Testing for Population Proportion

Hypothesis testing for population proportion for single and two population are discussed below.

4.1 Hypothesis Testing for Single Population Proportions

The hypothesis testing of mean is mostly used but, in many situations, especially in business, social sciences, engineering and behavioural sciences, we would like to test a hypothesis regarding the population proportion of the number of observations/units/ elements that possess a certain attribute instead of testing the population mean. For example:

- One may be interested in hypothesis testing that 56% of the adult people of a city use Facebook,
- A medicine research team of an institute wants to test whether a new treatment cures a disease more than 85%,
- A market researcher wants to test whether the proportion of new car purchasers who are female has been increased significantly,
- The course coordinator of the MST-016 course wants to test the hypothesis that 75% of the students are satisfied with the counselling classes of the course,
- A student of the MSCAST programme is interested in testing whether the passing ratio of the students of the MSCAST programme is greater than 50%, etc.

To test the hypotheses in such situations, we require tests for population proportion. The procedure of hypothesis testing for the population proportion has a similar rationale and approach as the mean.

Suppose we want to test the hypothesis that the population proportion (P) has a specified value, say, P0. For the hypothesis testing procedure, first of all, we set up the null and alternative hypotheses. Here, we want to test the claim about the specified value P0 of population proportion P so we can formulate the null and alternative hypotheses as

$H_0: P = P_0$ and $H_1 : P \neq P_0$ [two tailed test]

Or

$H_0: P \leq P_0$ and $H_1 : P > P_0$ [right- tailed test]

Or

$H_0: P \geq P_0$ and $H_1 : P < P_0$ [left- tailed test]

For testing the hypothesis, suppose we take a random sample of size n from the population with population proportion P. Also, suppose X denote the number of observations/units/ elements that possess a certain attribute (successes) out of n observations of the sample then we can define the sample proportion p as

$$P=\frac{X}{n} \leq 1$$

The tests for the population mean depend on the sampling distribution of the sample mean, in a similar way, the tests for population proportion also depend on the sampling distribution of the sample proportion. The sampling distribution of the proportion converges to the normal distribution with mean P and variance P(1 – P)/n if

- nP> 15 and
- n(1-P)>15

whatever the form of the population may be. Therefore, we consider the following test statistic for hypothesis testing for a population proportion:

$$z = \frac{p-P}{\sqrt{P(1-P)/n}} \sim N(0,1)$$

Since the test statistic is calculated under the assumption that the null hypothesis is true and here, the null hypothesis states that $P = P_0$ or $P \leq P_0$ or $P \geq P_0$. In each case, we take $P = P_0$ under the null hypothesis, therefore, the test statistic becomes as

$$z = \frac{p-P_0}{\sqrt{P_0(1-P_0)/n}}$$ under null hypothesis.

The test statistic follows the standard normal distribution with mean 0 and variance 1. Therefore, if nP >15 and n(1 – P) > 15 then we apply the Z-test for hypothesis testing for a single population proportion.

After deciding the test, we follow the rest of the steps of the procedure of hypothesis testing of the Z-test for a single population mean as discussed. That is, we calculate the value of the test statistic based on the observed sample and compare it with the critical (or cut-off) value(s) in the sampling distribution of the test statistic. If the test statistic lies within the. If the test statistic lies within the rejection region of the null hypothesis, then we reject the null hypothesis otherwise we do not reject it at the given level of significance. Finally, we draw a conclusion about the claim of the population proportion.

p-Value Approach

We calculate the p-value using the Z-test for hypothesis testing of a population proportion and take the decision of rejection or non-rejection of the null hypothesis using the p-value approach as discussed.

Solved Problem 1: A manager of a fast-food chain observed that they had served the order correctly 85% of the time. To improve it, the manager has developed a new process to make sure that orders are served correctly. To test the new process, he checked a sample of 200 orders using the new process and observed 182 were served correctly. To test whether the new process has increased the proportion of orders served correctly or not:

(i) Formulate the null and alternative hypothesis for this test situation.

(ii) Which test, one-tailed or two-tailed is used and why?

(iii) Conduct a hypothesis test for this situation at the 5% level of significance.

(iv) What management conclusion would be drawn from the findings?

(v) What conclusion could be drawn from the p-value?

Solution: Here, we are given that

The specified value of population proportion (P_0) = 0.85 The sample size (n) = 200

The number of times the order is correctly served (X) = 182

We can calculate the sample proportion of the order correctly served as

$$\bar{p} = \frac{X}{n} = \frac{182}{200} = 0.91$$

Here, we want to test that the new process has increased the proportion of orders served correctly. We can say that the new process has increased the proportion of orders served correctly if it is greater than 85% of the time. If we denote the proportion of orders served correctly by P, then the claim is P > 0.85 and its complement is P ≤ 0.85. Therefore, we can formulate the null and alternative hypothesis as

H_0 : P ≤ 0.85 [the proportion of orders served correctly has not increased]

H_1 : P > 0.85 [the proportion of orders served correctly has increased] (claim)

Since the alternative hypothesis is right-tailed, therefore, we will use a right-tailed test.

Now, you have to decide which test is appropriate for this situation. To decide which test out of these will be used, you have to check whether the conditions of normality are fulfilled or not as

nP = 200 × 0.85 = 170 > 15 and

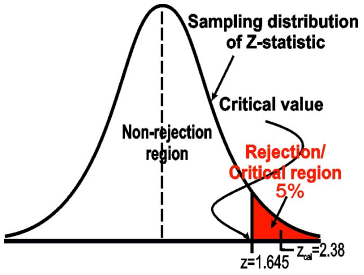
n(1- P) = 200 × (1- 0.85) = 200 × 0.15 = 30 > 15

Since the conditions of the normality are fulfilled so we go for the Z-test and the test statistic for the Z-test for a single population proportion is given by

$$z = \frac{p - P_0}{\sqrt{P_0(1-P_0)/n}}$$

We can calculate the value of the test statistic using the sample as

$$z = \frac{0.91 - 0.85}{\sqrt{0.85(1-0.85)/200}} = \frac{0.06}{0.0252} = 2.38$$



Since the alternative hypothesis is right-tailed, therefore, the critical region will lie at the right-tailed of the distribution as shown in the above figure. From the Z-table (Table I), the critical value for the right-tailed test at α= 0.05 is $z_\alpha = z_{0.05}$ =1.645

Since the calculated value of the test statistic (= 2.38) is greater than the critical value (=1.645), it means that it lies in the rejection region at the 5% level of significance as shown in the above figure. So, we reject the null hypothesis and support the alternative hypothesis. Since the alternative hypothesis is the claim so we support the claim.

Thus, we conclude that the sample does not provide us with sufficient evidence against the manager's belief so we may assume that the new process has increased the proportion of orders served correctly.

Decision using p-value approach:

Since the test is right-tailed, therefore, we can calculate the p-value as

p-value=P [$Z \geq z$]=P [$Z \geq 2.38$]

=1-P [$Z \leq 2.38$]

=1-0.9913=0.0087.

Since the p-value (=0.0087) is less than α (=0.05). Therefore, we reject the null hypothesis at the 5% level of significance. Therefore, we draw the same conclusion as we have obtained using the critical value approach.

Solved Problem 2: Online grocery company believes that more than 78% of the visitors to its website will make a purchase. A researcher in the marketing department thinks to test this claim. The researcher randomly selected 157 visits to the website and found that 126 of the visits resulted in a purchase. Does the company's belief that more than 78% of the visitors to its website will make a purchase at a 5% level of significance?

Solution:

Step 1: State the Hypotheses

The company believes that **more than 78%** of the visitors make a purchase.

$$H_0: p = 0.78$$
$$H_1: p > 0.78$$

This is a **right-tailed test**.

Step 2: Check the Conditions

Since

$$nP_0 = 157 \times (0.78) = 122.46 > 5$$

and

$$n(1 - P_0) = 157 \times (0.22) = 34.54 > 5,$$

the normal approximation is appropriate.

Step 3: Compute the Sample Proportion

$$\hat{p} = \frac{x}{n} = \frac{126}{157} = 0.8025.$$

Step 4: Compute the Test Statistic

The test statistic is

$$z = \frac{\hat{p} - P_0}{\sqrt{\frac{P_0(1 - P_0)}{n}}}.$$

Substituting the given values,

$$z = \frac{0.8025 - 0.78}{\sqrt{\frac{0.78(0.22)}{157}}} = \frac{0.0225}{\sqrt{0.001093}} = \frac{0.0225}{0.0331} = 0.68.$$

Step 5: Calculate the p-value

For a right-tailed test,

$$\text{p-value} = P(Z > 0.68).$$

From the standard normal table,

$$P(Z < 0.68) = 0.7517.$$

Hence,

$$\text{p-value} = 1 - 0.7517 = 0.2483.$$

Step 6: Decision Rule

Compare the p-value with the significance level.

- p-value $= 0.2483$
- $\alpha = 0.05$

Since

$$0.2483 > 0.05,$$

Do not reject the null hypothesis.

Step 7: Conclusion

At the **5% level of significance**, there is **insufficient evidence** to conclude that more than **78%** of the visitors to the website make a purchase.

4.2 Hypothesis Testing for Comparing two Population Proportions

In the previous subsection, you studied which test is applied for hypothesis testing for a population proportion. There are many situations where someone is interested in comparing the proportions of an attribute in two different populations/ groups. For example:

- A market researcher wants to test whether the proportion of men who shop for bargains is significantly lower than the proportion of women who shop for bargains,
- A researcher wants to test whether there is a significant difference between the proportions of taxes on the citizens of two countries,
- An analyst wishes to determine whether the proportion of consumption of tea is more than coffee in a city, etc.

In each example, someone is interested in comparing the proportions of an attribute in two different populations/groups. For testing such types of claims/statements, we need to study the procedure and tests for hypothesis testing for comparing the proportions of an attribute in two different populations/groups. Let us discuss it.

Suppose we are interested in hypothesis testing for comparing the proportions of an attribute in two different populations, say, Population-I and Population-II and P_1 and P_2, denote their proportions of an attribute/characteristic, respectively. To test the hypothesis about the comparison of two population proportions of an attribute in two populations, we can formulate the null and alternative hypotheses as

$$H_0: P_1 = P_2 \quad \text{and} \quad H_1: P_1 \neq P_2 \quad \text{[two tailed test]}$$

Or

$$H_0: P_1 \leq P_2 \quad \text{and} \quad H_1: P_1 > P_2 \quad \text{[right- tailed test]}$$

Or

$$H_0: P_1 \geq P_2 \quad \text{and} \quad H_1: P_1 < P_2 \quad \text{[left- tailed test]}$$

To test the hypothesis, suppose we draw two independent random samples of sizes n_1 and n_2 from Population-I and Population-II, respectively and also, suppose X_1 and X_2 represent the number of observations or elements possessing the attribute under study, respectively. We can calculate the sample proportions p_1 and p_2 as

$$p_1 = \frac{X_1}{n_1} \quad \text{and} \quad p_2 = \frac{X_2}{n_2}$$

As you have seen the tests for the single population proportion depend on the sampling distribution of the sample proportion, in a similar way, the tests for comparing proportions of two populations depend on the sampling distribution of the difference of two sample proportions, that is, sampling distribution of $(p_1 - p_2)$. The sampling distribution of difference of two proportions converges to the normal distribution with mean $(P_1\text{-}P_2)$ and variance $\frac{P_1(1-P_1)}{n_1} + \frac{P_2(1-P_2)}{n_2}$ under the normality conditions

- $n_1P_1 > 15$, $n_1(1-P_1) > 15$ and
- $n_2P_2 > 15$, $n_2(1-P_2) > 15$

Note: Generally, the population proportions P1 and P2 are unknown, so we take sample proportions p1 and p2 in the conditions of normality, therefore,

- $n_1p_1 > 15$, $n_1(1-p_1) > 15$ and
- $n_2p_2 > 15$, $n_2(1-p_2) > 15$

Thus, we consider the following test statistic for hypothesis testing for comparing two population proportions:

$$z=\frac{p_1-p_2-(P_1-P_2)}{\sqrt{\frac{P_1(1-P_1)}{n_1}+\frac{P_2(1-P_2)}{n_2}}} \sim N(0,1)$$

Since the test statistic is calculated under the assumption that the null hypothesis is true and the null hypothesis states that $P_1 = P_2$ or $P_1 \geq P_2$ or $P_1 \leq P_2$. In each case, we take

$$P_1 = P_2 = P \Rightarrow P_1 - P_2 = 0$$

under the null hypothesis, therefore the test statistic becomes as

$$z=\frac{p_1-p_2}{\sqrt{\frac{P(1-P)}{n_1}+\frac{P(1-P)}{n_2}}}$$ under null hypothesis

Where P is estimated by $\hat{P}$ using sample information as

$$\hat{P} = \frac{n_1p_1+n_2p_2}{n_1+n_2} = \frac{X_1+X_2}{n_1+n_2}$$

The test statistic approximately follows the standard normal distribution with mean 0 and variance 1. Therefore, if the normality conditions are fulfilled then we apply the Z-test for comparing two proportions.

After deciding an appropriate test, we follow the rest of the steps of the procedure of testing of hypothesis for a single proportion discussed in the previous section.

Let us take an example to show how to apply the Z-test to compare the proportions of two independent groups in real-life situations.

Solved Problem 1: A market researcher believes that men who shop for bargains are less likely than women. To test his belief, he conducted a survey and observed that 143 men out of 352 men go for bargains whereas 209 women out of 417 women go for bargains. Test whether the proportion of men who shop for bargains is significantly less than the proportion of women who shop for bargains at a 5% level of significance.

Solution: Here, we are given that

n_1 = total men in the survey = 352, X_1 = the men who bargains = 143

n_2 = total women in the survey = 417, X_2 = the women who bargains = 209

We can calculate the sample proportion of men who shop for bargains as

$$p_1 = \frac{X_1}{n_1} = \frac{143}{352} = 0.41$$

and the sample proportion of women who shop for bargains as

$$p_2 = \frac{X_2}{n_2} = \frac{209}{417} = 0.50$$

Here, we want to test that the proportion of men who shop for bargains is significantly lower than the proportion of women who shop for bargains. If the proportions of men and women who shop for bargains are denoted by P_1 and P_2, respectively, then the claim is $P_1 < P_2$ and its complement is $P_1 \geq P_2$. Since the complement contains the equality sign so we can take the complement as the null hypothesis and the claim as the alternative hypothesis. Therefore, we can formulate the null and alternative hypotheses as

H_0: $P_1 \geq P_2$ [the proportion of men who shop for bargains is not less than to women]

H_1: $P_1 < P_2$ [the proportion of men who shop for bargains is less than to women] (claim)

Since the alternative hypothesis is left-tailed, therefore, we will use a left-tailed test. To decide which test out of these will be used, you have to check whether the conditions of normality are fulfilled or not as

$n_1 p_1 = 352 \times 0.41 = 144.32 > 15$

$n_1 (1 - p_1) = 352 \times (1 - 0.41) = 207.68 > 15$

$n_2 p_2 = 417 \times 0.5 = 208.5 > 15$

$n_2(1 - p_2) = 417 \times (1 - 0.5) = 208.5 > 15$

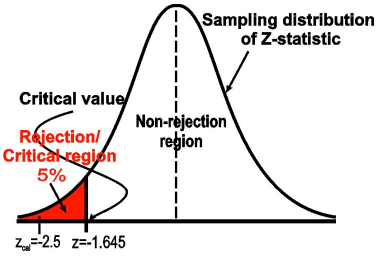
Since the conditions of the normality are fulfilled so we go for the Z-test and the test statistic for the Z-test for comparing two population proportions is given by

$$Z = \frac{p_1 - p_2}{\sqrt{\frac{P(1-P)}{n_1} + \frac{P(1-P)}{n_2}}}$$

Where P is estimated by $\hat{P}$ using sample information as

$$\hat{P} = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2} = \frac{X_1 + X_2}{n_1 + n_2} = \frac{143 + 2}{352 + 4} = 0.46.$$

We can estimate the combined proportion (P) as



$$Z=\frac{p_1-p_2}{\sqrt{\frac{P(1-P)}{n_1}+\frac{P(1-P)}{n_2}}}=\frac{0.41-0.50}{\sqrt{0.46(1-0.46)(\frac{1}{352}+\frac{1}{417})}}=-2.5$$

Since the alternative hypothesis is left-tailed, therefore, the critical region will lie at the left-tailed of the distribution as shown in the above figure. Thus, from the Z-table (Table I), the critical value for the left-tailed test at α = 0.05 is

$$-z_\alpha = -z_{0.05} = -1.645.$$

Since the calculated value of the test statistic (= –2.5) is less than the critical value (= –1.645), it means that it lies in the rejection region at the 5% level of significance as shown in Fig. 17.4. So, we reject the null hypothesis and support the alternative hypothesis. Since the alternative hypothesis is the claim so we support the claim. Thus, we conclude that the samples do not provide us with sufficient evidence against the researcher's belief so we may assume that the proportion of men who shop for bargains is significantly lower than the proportion of women who shop for bargains at the 5% level of significance.

Solved Problem 2: It is found that 250 errors in the randomly selected 1000 lines of code from Team A and 300 errors in 800 lines of code from Team B. Can we assume that team B's performance is superior to that of A.

Solution:

Step 1: State the Hypotheses

$$H_0: p_1 = p_2$$
$$H_1: p_2 < p_1$$

where

- p_1= proportion of errors for Team A,
- p_2= proportion of errors for Team B.

This is a left-tailed test.

Step 2: Compute the Pooled Proportion

$$\hat{p} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{250 + 300}{1000 + 800} = \frac{550}{1800} = 0.3056.$$

Therefore,

$$1 - \hat{p} = 0.6944.$$

Step 3: Compute the Test Statistic

The test statistic is

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}.$$

Substituting the values,

$$z = \frac{0.25 - 0.375}{\sqrt{(0.3056)(0.6944)\left(\frac{1}{1000} + \frac{1}{800}\right)}}$$
$$= \frac{-0.125}{\sqrt{0.2122(0.00225)}}$$
$$= \frac{-0.125}{0.02185} = -5.72.$$

Step 4: Calculate the p-value

For a left-tailed test,

Learn more

$$\text{p-value} = P(Z < -5.72).$$

From the standard normal distribution,

$$\boxed{\text{p-value} \approx 0.0000.}$$

Step 5: Decision Rule

Assume the significance level is

$$\alpha = 0.05.$$

Since

$$\text{p-value} = 0.0000 < 0.05,$$

we reject the null hypothesis.

Step 6: Conclusion

The sample error rate for Team B is

$$37.5\%,$$

which is significantly higher than Team A's error rate of

$$25\%.$$

Hence, Team B makes significantly more errors than Team A.

Final Answer

Since the p-value is less than 0.05, we reject the null hypothesis. There is sufficient statistical evidence to conclude that Team B's performance is not superior to Team A's. In fact, Team B has a significantly higher error rate, indicating that Team A performs better.

Practice Problem Set

Problem 1: A medical researcher speculates that the proportion of smokers who have wrinkles around the eyes is greater than non-smokers. In an observational study, the researcher enrolled 120 smokers and 230 non-smokers. Based on a standardised wrinkle score, the researcher discovered that 85 of the smokers and 134 of the non-smokers had noticeable wrinkles around their eyes in the sample. Test the belief of the medical researcher at a 1% level of significance.

Problem 2: A Telecom service provider claims that individual customers pay on an average 400 rs. per month with standard deviation of 25 rs. A random sample of 50 customers bills during a given month is taken with a mean of 250 and standard deviation of 15. What to say with respect to the claim made by the service provider?

Problem 3: From the data available, it is observed that 400 out of 850 customers purchased the groceries online. Can we say that most of the customers are moving towards online shopping even for groceries?

Problem 4: A company manufacturing automobile tyres finds that tyre-life is normally distributed with a mean of 40,000 km and a standard deviation of 3000 km. It is believed that a change in the production process will result in a better product and the company has developed a new tyre. A sample of 100 new tyres has been selected. The company has found that the mean life of these new tyres is 40,900 km. Can it be concluded that the new tyre is significantly better than the old one, using the significant level of 0.01?

Problem 5: The mortgage department of a large bank is interested in the nature of loans of first-time borrowers. This information will be used to tailor their marketing strategy. They believe that 50% of first-time borrowers take out smaller loans than other borrowers. They perform a hypothesis test to determine if the percentage is the same or different from 50%. They sample 100 first-time borrowers and find 53 of these loans are smaller that the other borrowers. For the hypothesis test, they choose a 5% level of significance.

Problem 6: Suppose a consumer group suspects that the proportion of households that have three or more cell phones is 30%. A cell phone company has reason to believe that the proportion is not 30%. Before they start a big advertising campaign, they conduct a hypothesis test. Their marketing people survey 150 households with the result that 43 of the households have three or more cell phones.

Problem 7: A smartphone manufacturer claims the average battery life of their new model is 20 hours. You suspect it is actually less. You test a sample of 35 phones and find a sample mean of 19.2 hours. Assume the population standard deviation (σ) is 2.5 hours. Test the claim at a 5% significance level ($\alpha = 0.05$).

5. Testing of hypothesis about mean

In this section, we formally consider tests of hypotheses on a single population mean.

5.1 Test on a Single Mean (Variance Known)

We should first describe the assumptions on which the experiment is based. The model for the underlying situation centers around an experiment with $X_1, X_2, \ldots, X_n$ representing a random sample from a distribution with mean μ and variance $\sigma^2 > 0$

Consider first the hypothesis

$$H_0 : \mu = \mu_0$$

$$H_1 : \mu \neq \mu_0$$

The appropriate test statistic should be based on the random variable X. The Central Limit Theorem essentially states that despite the distribution of X, the random variable $\bar{X}$ has approximately a normal distribution with mean μ and variance σ^2/n for reasonably large sample sizes. So, $\mu_{\bar{X}} = \mu$ and $\sigma^2_{\bar{X}} = \sigma^2/n$. We can then determine a critical region based on the computed sample average $\bar{x}$. It should be clear by now that there will be a two-tailed critical region for the test.

Standardization of $\bar{X}$:

It is convenient to standardize random variable Z, where

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

Note that, under H_0, that is, if $\mu = \mu_0$ then, $\sqrt{n}(\bar{x} - \mu_0/\sigma)$ follows an $n(x; 0, 1)$ distribution, and hence the expression

$$P(-z_{\alpha/2} < \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} < z_{\alpha/2}) = 1 - \alpha$$

can be used to write an appropriate non - rejection region. The reader should keep in mind that, formally, the critical region is designed to control α, the probability of type I error. It should be obvious that a two-tailed signal of evidence is needed to support H_1. Thus, given a computed value $\bar{x}$, the formal test involves rejecting H_0 if the computed test statistic z falls in the critical region described next.

Test Procedure for a Single Mean (Variance Known):

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} > z_{\alpha/2} \quad or \quad z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} < -z_{\alpha/2}$$

- If $-z_{\alpha/2} < z < z_{\alpha/2}$, do not reject H_0.
- Rejection of H_0, of course, implies acceptance of the alternative hypothesis $\mu \neq \mu_0$.
- It should be clear that there will be probability α of rejecting H_0 (falling into the critical region) when, indeed, $\mu = \mu_0$

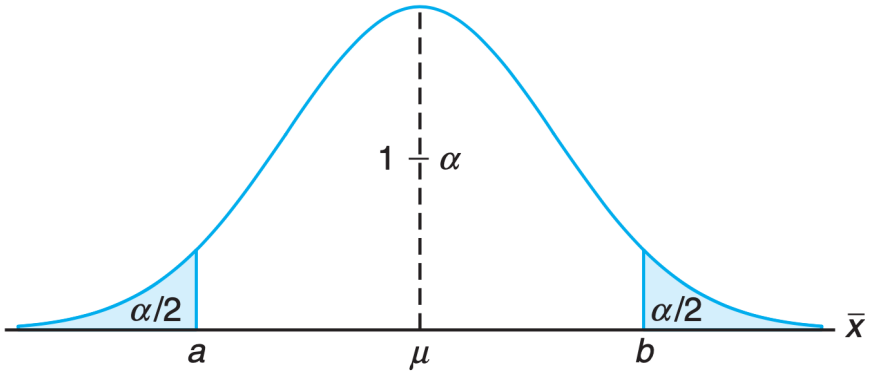


Figure 1: Critical region for the alternative hypothesis $\mu = \mu_0$

Hence, for a significance level α the critical values of the random variable z and $\bar{x}$ are both depicted in Fig. 1

Tests of one-sided hypotheses on the mean involve the same statistic described in the two-sided case. The difference, of course, is that the critical region is only in one tail of the standard normal distribution. For example, suppose that we seek to test

$$H_0 : \mu = \mu_0$$

$$H_1 : \mu > \mu_0$$

- The signal that favors H_1 comes from large values of z.
- Rejection of H_0 results when the computed $z > z_\alpha$.
- If the alternative is $H_1 : \mu < \mu_0$, the critical region is entirely in the lower tail and thus rejection results from $z < -z_\alpha$.
- In a one-sided testing case the null hypothesis can be written as $H_0 : \mu \leq \mu_0$ or $H_0 : \mu \geq \mu_0$, it is usually written as $H_0 : \mu = \mu_0$.

Example 0.0.1 *A random sample of 100 recorded deaths in the United States during the past year showed an average life span of 71.8 years. Assuming a population standard deviation of 8.9 years, does this seem to indicate that the mean life span today is greater than 70 years? Use a 0.05 level of significance.*

Sol:

1. $H_0 : \mu = 70\ years$
2. $H_1 : \mu > 70\ years$
3. $\alpha = 0.05$
4. *Critical region:* $z > 1.645$, *where* $z = \frac{\bar{x}-\mu_0}{\sigma/\sqrt{n}}$

5. *Computations:* $\bar{x} = 71.8$ *years,* $\sigma = 8.9$ *years, and hence* $z = \frac{\bar{x}-\mu_0}{\sigma/\sqrt{n}} = 71.8 - 70\ 9/\sqrt{1}00 = 2.02$.

6. *Decision: Reject* H_0 *and conclude that the mean life span today is greater than 70 years.*

The P-value corresponding to $z = 2.02$ *is given by the area of the shaded region in Fig.2(a)*

Now, $P = P(Z > 2.02) = 0.0217$ *As a result, the evidence in favor of* H_1 *is even stronger than that suggested by a 0.05 level of significance.*

Example 0.0.2 *A manufacturer of sports equipment has developed a new synthetic fishing line that the company claims has a mean breaking strength of 8 kilograms with a standard deviation of 0.5 kilogram. Test the hypothesis that* $\mu = 8$ *kilograms against the alternative that* $\mu = 8$ *kilograms if a random sample of 50 lines is tested and found to have a mean breaking strength of 7.8 kilograms. Use a 0.01 level of significance.*

Sol:

1. $H_0 : \mu = 8\ kilograms$

2. $H_1 : \mu \neq 8\ kilograms$

3. $\alpha = 0.01$

4. *Critical region:* $z < -2.575$ *and* $z > 2.575$ *where,* $z = \frac{\bar{x}-\mu_0}{\sigma/\sqrt{n}}$

5. *Computations:* $\bar{x} = 7.8$ *kilograms,* $\sigma = 0.5$ *kilogram, and hence* $z = \frac{\bar{x}-\mu_0}{\sigma/\sqrt{n}} = 7.8 - 8\ 0.5/\sqrt{5}0 = -2.83$.

6. *Decision: Reject* H_0 *and conclude that the average breaking strength is not equal to 8 but is, in fact, less than 8 kilograms.*

Since the test in this example is two tailed, the desired P-value is twice the area of the shaded region in Fig.2(b) to the left of $z = 2.83$. *Therefore, using Table we have*

$$P = P(|Z| > 2.83) = 2P(Z < -2.83) = 0.0046$$

which allows us to reject the null hypothesis that $\mu = 8$ *kilograms at a level of significance smaller than 0.01.*

5.2 Test on a Single Sample (Variance Unknown)

In this section we consider tests for the population mean assuming that the population variance is unknown. We shall thus invoke Student t-distribution. The random variables $X_1, X_2, \ldots, X_n$ represent a random sample from a normal distribution with unknown mean μ and variance $\sigma^2 > 0$. Then the random variable $\sqrt{n}(\bar{x} - \mu)/s$ has a Student t - distribution with n -1 degrees of freedom.

The t-Statistic for a Test on a Single Mean (Variance Unknown)

For the two-sided hypothesis

$$H_0 : \mu = \mu_0$$

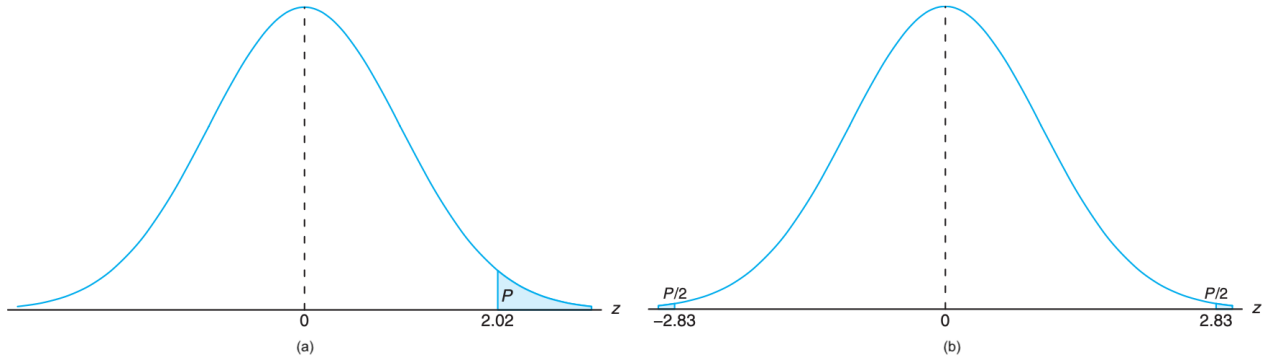


Figure 2: P values of Problems 1 and 2

$$H_1 : \mu \neq \mu_0$$

we reject H_0 at significance level α when the computed t-statistic

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

exceeds $t_{\alpha/2,n-1}$ or is less than $-t_{\alpha/2,n-1}$.

- The t-distribution is symmetric around the value zero. Thus, this two-tailed critical region applies in a fashion similar to that for the case of known σ.
- For the two-sided hypothesis at significance level α, the two-tailed critical regions apply.
- For $H_1 : \mu > \mu_0$, rejection results when $t > t_{\alpha,n-1}$.
- For $H_1 : \mu < \mu_0$, the critical region is given by $t < -t_{\alpha,n-1}$.

Example 0.0.3 *The Edison Electric Institute has published figures on the number of kilowatt hours used annually by various home appliances. It is claimed that a vacuum cleaner uses an average of 46 kilowatt hours per year. If a random sample of 12 homes included in a planned study indicates that vacuum cleaners use an average of 42 kilowatt hours per year with a standard deviation of 11.9 kilowatt hours, does this suggest at the 0.05 level of significance that vacuum cleaners use, on average, less than 46 kilowatt hours annually? Assume the population of kilowatt hours to be normal.*

Sol:

1. $H_0 : \mu = 46$ *kilowatt hours*
2. $H_1 : \mu < 46$ *kilowatt hours*
3. $\alpha = 0.05$
4. *Critical region:* $t < -1.796$, *where* $t = \frac{\bar{x}-\mu_0}{s/\sqrt{n}}$ *with 11 degrees of freedom*

5. *Computations:* $\bar{x} = 42$ *kilowatt hours,* $s = 11.9$ *kilowatt hours and* $n = 12$.

   Hence, $t = \frac{42-46}{11.9/\sqrt{12}} = -1.16.$, $P = P(T < -1.16) \approx 0.135$

6. *Decision: Do not eject* H_0 *and conclude that the average number of kilowatt hours used annually by home vacuum cleaners is not significantly less than 46.*

Example 0.0.4 *It is assumed that the mean systolic blood pressure in* $\mu = 120$ *mm Hg. In the Honolulu Herat Study, a sample of* $n = 100$ *people had an average systolic blood pressure of 130.1 mm Hg with a standard deviation of 21.21 mm Hg. Test whether the group is significantly different from the regular population. You may assume* $t_{0.025,99} = 1.9842$

Sol: *Let* μ *be the population mean systolic blood pressure, assumed to be* $\mu = 120$ *mm Hg.*

The data is of a given sample of $n = 100$ *people with an average systolic blood pressure of* $\bar{x} = 130.1$ *mm Hg with a standard deviation of* $s = 21.21$ *mm Hg.*

1. $H_0 : \mu = 120$ *mm Hg*
2. $H_1 : \mu \neq 120$ *mm Hg*
3. $\alpha = 0.05$
4. *Critical value:* $t_{0.025,99} = 1.9842$, *where* $t = \frac{\bar{x}-\mu_0}{s/\sqrt{n}}$ *with* $n - 1 = 100 - 1 = 99$ *degrees of freedom*
5. *Substituting the values,*

   $t = \frac{130.1-120}{21.21/\sqrt{100}} \approx 4.76.$,
6. *For* $t = 4.76$ *with 99 degrees of freedom, P - value* < 0.0001.

   This P-value is much smaller than the usual significance level of 0.05 that is, P-value < 0.05
7. *Decision: We therefore reject the null hypothesis* H_0 *and conclude that the population mean systolic blood pressure is significantly different from 120 mm Hg.*

6. Two-sample tests

A two-sample test is a statistical hypothesis test used to determine whether two independent population parameters (usually means or proportions) are significantly different based on information obtained from two independent random samples.

Unlike a one-sample test, which compares a sample statistic with a known population parameter, a two-sample test compares two sample statistics to draw conclusions about their corresponding populations.

Why Do We Need a Two-Sample Test?

In practice, researchers often wish to compare two groups rather than compare one group with a known standard.

Examples include:

- Comparing the average marks of students taught by two different teachers.
- Comparing the average lifespan of bulbs manufactured by two factories.
- Comparing the average recovery times of patients receiving two different medicines.
- Comparing the average crop yields obtained using two fertilizers.

6.1 Test for equality of variance

The F-test is performed to test the equality of two population variances.

Why is the F-test Used?

The F-test is used because, if two independent random samples are drawn from normally distributed populations, the ratio of their sample variances follows an F-distribution under the null hypothesis that the population variances are equal.

Mathematically, if $H_0 : {\sigma_1}^2 = {\sigma_2}^2$

then the statistic $F = \frac{S_1^2}{S_2^2}$

where, S_1^2 is the larger sample variance and S_2^2 is the smaller sample variance, follows an F-distribution with

- Numerator degrees of freedom: $d_1 = n_1 - 1$
- Denominator degrees of freedom: $d_2 = n_2 - 1$

This property allows us to determine whether the observed difference between the sample variances is statistically significant.

Definition 0.1 *The F - distribution (see Fig.3) with d_1 and d_2 degrees of freedom is the distribution of*

$$X = \frac{U_1/d_1}{U_2/d_2}$$

where U_1 and U_2 are independent random variables with chi-square distributions with respective degrees of freedom d_1 and d_2.

Test Procedure for the F-Test

1. Null Hypothesis: $H_0 : {\sigma_1}^2 = {\sigma_2}^2$

   Alternative Hypothesis:

   Depending on the problem, one of the following alternatives is used:

   Two-tailed Test: $H_0 : {\sigma_1}^2 \neq {\sigma_2}^2$

   Right-tailed Test: $H_0 : {\sigma_1}^2 > {\sigma_2}^2$

   Left-tailed Test: $H_0 : {\sigma_1}^2 < {\sigma_2}^2$

2. Choose the Level of Significance (α)

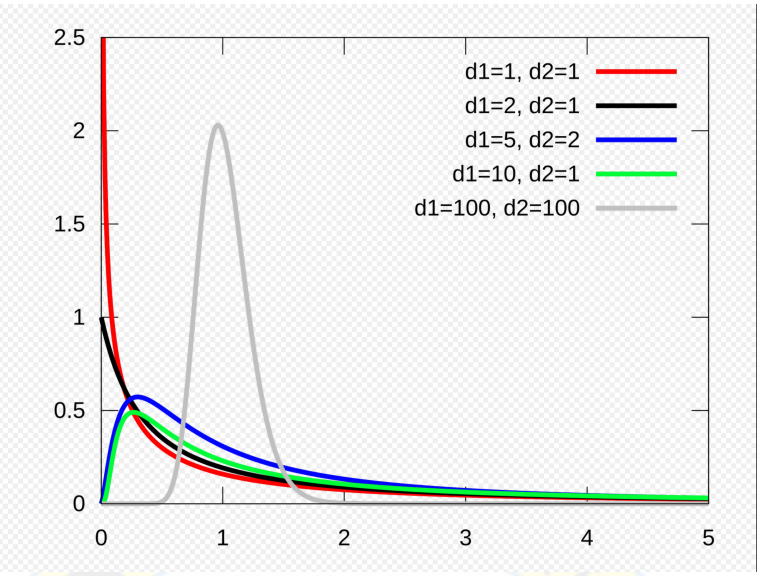


Figure 3: Probability density function of F-distribution

3. Calculate the Sample Variances: If raw data are given compute:

   $s_1^2 = \frac{\Sigma(x_1 - \bar{x}_1)^2}{n_1 - 1}$ and $s_2^2 = \frac{\Sigma(x_2 - \bar{x}_2)^2}{n_2 - 1}$

   [If s_1^2 and s_2^2 are already provided, this step is not necessary.]

4. Compute the F-Test Statistic:

   Place the larger sample variance in the numerator so that the test statistic is at least 1.

$$F = \frac{LargerSampleVariance}{SmallerSampleVariance} = \frac{s_1^2}{s_2^2}$$

   where, $s_1^2 > s_2^2$.

5. Determine the Degrees of Freedom: Compute

   - Numerator degrees of freedom: $d_1 = n_1 - 1$
   - Denominator degrees of freedom: $d_2 = n_2 - 1$

6. Find the Critical F-Value: Using the chosen significance level α and the degrees of freedom (d_1, d_2), obtain the critical value $F_\alpha(d_1, d_2)$ from the F-distribution table.

7. Make the Decision: If $F_{calculated} > F_{critical}$, reject the null hypothesis, else do not reject the null hypothesis.

8. Draw the Conclusion: Interpret the result in the context of the problem.

   - **Reject H_0:** There is sufficient evidence to conclude that the two population variances are significantly different.

- **Fail to reject H_0:** There is insufficient evidence to conclude that the population variances differ; they may be regarded as equal.

Example 0.1.1 *Two random samples were drawn from two normal populations and their values are:*

A	65	66	73	80	82	84	88	90	92		
B	64	66	74	78	82	85	87	92	93	95	97

Test whether the two populations have the same variance at the 5 percent level of significance. (Given: $F = 3.36$ at 5 percent level of significance for $d_1 = 10$ and $d_2 = 8$.)

Sol: *Null Hypothesis:* $H_0 : {\sigma_1}^2 = {\sigma_2}^2$

Alternative Hypothesis: $H_0 : {\sigma_1}^2 \neq {\sigma_2}^2$

A X_1	$(X_1-\bar{X}_1)$ x_1	x_1^2	B X_2	$(X_2-\bar{X}_2)$ x_2	x_2^2
65	-15	225	64	-19	361
66	-14	196	66	-17	289
73	-7	49	74	-9	81
80	0	0	78	-5	25
82	2	4	82	-1	1
84	4	16	85	2	4
88	8	64	87	4	16
90	10	100	92	9	81
92	12	144	93	10	100
			95	12	144
			97	14	196
$\sum X_1$= 720	$\sum x_1$=0	$\sum x_1^2$-=798	$\sum X_2$=913	$\sum x_2$=0	$\sum x_2^2$=1298

Figure 4: Table for computation of values

(For calculation refer to Table shown in Fig.4)

$\bar{x_1} = \frac{\sum x_1}{n_1} = 720/9 = 80$

${s_A}^2 = 798/9 - 1 = 99.75$

$\bar{x_2} = \frac{\sum x_2}{n_2} = 913/11 = 83$

${s_B}^2 = 1298/11 - 1 = 129.80$

$F = \frac{{s_B}^2}{{s_A}^2} = 1.3013$

At 5 percent level of significance, for $d_1 = 10$ *and* $d_2 = 8$*, the value of* $F_{0.05} = 3.36$

Since, $1.3013 < 3.36$

Do not reject the null hypothesis.

There is no significant difference between the variances of the two populations. We conclude that they have the same variance at the 5 percent level of significance.

Example 0.1.2 *A quality control engineer wants to determine whether two machines produce items with the same variability in weight. Two independent random samples are selected: Machine A:*

$n_1 = 12$, *sample variance* $s_1^2 = 36$ *and Machine B:* $n_2 = 10$, *sample variance* $s_2^2 = 16$. *At 5 percent level of significance test whether the population variances of the two machines are equal. Given* $F_{0.025,(11,9)} = 4.03$

Sol: *Null Hypothesis:* $H_0 : {\sigma_1}^2 = {\sigma_2}^2$

Alternative Hypothesis: $H_0 : {\sigma_1}^2 \neq {\sigma_2}^2$

Level of significance $\alpha = 0.05 \implies \alpha/2 = 0.025$

Degrees of freedom are $d_1 = n_1 - 1 = 11$ *and* $d_2 = n_2 - 1 = 9$

$F = \frac{36}{16} = 2.25$

Given, $F_{0.025,(11,9)} = 4.03$ *and* $2.25 < 4.03$ *this implies that the calculated value does not fall in the critical region.*

Therefore, we fail to reject the Null hypothesis.

7. ANOVA (one-way analysis of variance)

In many practical situations, we need to compare the means of more than two populations. Performing several two-sample t-tests increases the probability of making an incorrect conclusion (Type I error). To overcome this problem, Analysis of Variance (ANOVA) is used.

One-Way ANOVA (One-Way Classification) is a statistical technique used to determine whether the means of three or more independent populations (or treatments) are equal.

What is One-Way Classification? In one-way classification, observations are classified according to one factor (or one criterion) only.

Examples:

- Comparing the average crop yield using four different fertilizers.
- Comparing the average marks of students taught by three teaching methods.
- Comparing recovery times under four different medicines.
- Comparing the fuel efficiency of cars manufactured by different companies.

Here, the factor (fertilizer, teaching method, medicine, company) is called the treatment.

Assumptions of One-Way ANOVA:

The following assumptions must be satisfied:

- Samples are randomly selected.
- Samples are independent.
- Each population is normally distributed.
- Population variances are equal (homogeneity of variance).

Hypotheses: Suppose there are k treatments.

Null Hypothesis

$$H_0 : \mu_1 = \mu_2 = \ldots = \mu_k$$

$$That\ is, all\ population\ means\ are\ equal.$$

Alternative Hypothesis

$$H_1 : At\ least\ one\ population\ mean\ differs$$

7.1 Testing equality of Population (Treatment) Means: One-way Classification

In one way classification, following steps are carrying out for computing F- ratio.

1. Firstly get the squared value of all the observation for different samples (column).
2. Get the sum total of sample observations as $\Sigma X_1, \Sigma X_2, \ldots, \Sigma X_k$ in each column.

3. Get the sum total of squared values for each column as $\Sigma {X_1}^2, \Sigma X_2^2, \ldots, \Sigma X_k^2$ in each column.
4. Finding the value of "T" by adding up all the sums of sample observations i.e. $T = \Sigma X_1 + \Sigma X_2 + \ldots + \Sigma X_k$
5. Compute the Correction Factor by the formula: $C.F. = \frac{T^2}{N}$
6. Find out Total sum of Squares (SST) through squared values and C F:
   $SST = (\Sigma {X_1}^2 + \Sigma X_2^2 + \ldots + \Sigma X_k^2) - C.F.$
7. Find out Sum of square between the samples SSC by following formula:
   $SSC = (\Sigma {X_1}^2/n_1 + \Sigma X_2^2/n_2 + \ldots + \Sigma X_k^2/n_k) - C.F.$
8. Finally, find out sum of squares within samples i.e. SSE as under: $SSE = SST - SSC$

Table 1: Analysis of Variance (ANOVA) Table

Source of Variation	Sum of squares (SS)	Degrees of freedom	Mean Squares (MS)	Variance ratio of F
Between samples	SSC	$df1 = C - 1$	$MSC = SSC/df1$	
Treatments	SSE	$df2 = N - C$	$MSE = SSE/df2$	
Within samples (errors)				
Total	SST	n -1		$F = \frac{MSC}{MSE}$

1. SSC = Sum of squares between samples (Columns)
2. SST = Total sum of the squares of Variations.
3. SSE = Sum of squares within the samples.
4. MSC = Mean sum of squares between samples
5. MSE = Mean sum of squares within samples

Example 0.1.3 *To test the significance of variation in the retail prices of a commodity in three principal cities, Mumbai, Kolkata, and Delhi, four shops were chosen at random in each city and the prices who lack confidence in their mathematical ability observed in rupees were as follows:*

Kanpur	15	7	11	13
Lucknow	14	10	10	6
Delhi	4	10	8	8

Do the data indicate that the price in the three cities are significantly different?

Sol: *Let us take the null hypothesis that there is no significant difference in the prices of a commodity in the three cities.*

Calculations for analysis of variance are shown in the Table in Fig. 5:

Sample 1		Sample 2		Sample 3	
Kanpur		Lucknow		Delhi	
x_1	x_1^2	x_2	x_2^2	x_3	x_3^2
15	225	14	196	4	16
7	49	10	100	10	100
11	121	10	100	8	64
13	169	6	36	8	64
$\sum x_1 = 46$	$\sum x_1^2$ =564	$\sum x_2$=40	$\sum x_2^2$ =432	$\sum x_3 = 30$	$\sum x_3^2$ =244

Figure 5: Calculations for ANOVA

There are $r =$ *treatments (samples) with* $n_1 = 4, n_2 = 4, n_3 = 4$ *and* $n = 12$

T= Sum of all the observations in the three samples

$= \Sigma X_1 + \Sigma X_2 + \Sigma X_3 = 46 + 40 + 30 = 116$

$C.F. = CorrectionFactor = \frac{T^2}{n} = \frac{116^2}{12} = 1121.33$

$SST =$ *Total sum of the squares* $= (\Sigma {X_1}^2 + \Sigma X_2^2 + \Sigma X_3^2) - C.F$

$= (564 + 432 + 244) - 1121.33 = 118.67$

$SSC =$ *Sum of the squares between the samples*

$= (\Sigma {X_1}^2/n_1 + \Sigma X_2^2/n_2 + \Sigma X_3^2/n_3) - C.F$

$= (46^2/4 + 40^2/4 + 30^2/4) - 1121.33 = 32.67$

$SSE = SST - SSC = 118.67 - 32.67 = 86$

Degrees of freedom: $df1 = r - 1 = 3 - 1 = 2$ *and* $df2 = n - r = 12 - 3 = 9$

Thus, $MSC = SSC/df1 = 32.67/2 = 16.33$ *and* $MSE = SSE/df2 = 86/9 = 9.55$

Table 2: Analysis of Variance (ANOVA) Table

Source of Variation	Sum of squares (SS)	Degrees of freedom	Mean Squares (MS)	Variance ratio of F
Between samples	SSC = 32.67	$df1 = 2$	$MSC = \frac{SSC}{df1} = \frac{32.67}{2} = 16.335$	$F = \frac{MSC}{MSE} = \frac{16.35}{9.55} = 1.71$
Within samples	SSE = 86	$df2 = 9$	$MSE = \frac{SSE}{df2} = \frac{86}{9} = 9.55$	
Total	SST = 118.67	n -1 = 11		

The table value of F for $df1 = 2$, $df2 = 9$ *and* $\alpha = 0.05$ *level of significance is 4.26.*

Analysis of Variance (ANOVA) Table is shown in Table 2 which gives $F = 1.71$

Since, calculated value of F is less than its critical (or table) value, the null hypothesis is accepted.

Hence we conclude that prices of a commodity in three cities have no significant difference.

Example 0.1.4 *A researcher wants to determine whether three different teaching methods produce different average scores in a mathematics test. Three groups of students are taught using different methods, and their scores are recorded as follows:*

Testing method A	Testing method B	Testing method C
70	80	90
75	85	95
80	90	85
85	75	100

At 5 percent level of significance, test whether there is a significant difference among the mean scores of the three teaching methods.

Sol: *Null hypothesis* $H_0 : \mu_A = \mu_B = \mu_C$

(All three teaching methods have the same population mean score.)

H_1 *: All three teaching methods have the same population mean score*

Number of groups $= 3$ *and observations in each group* $= 4$.

Therefore, total observations $N = 12$.

$T =$ *Sum of all the observations in the three samples*

$= \Sigma X_1 + \Sigma X_2 + \Sigma X_3 = 310 + 330 + 370 = 1010$

$C.F. = CorrectionFactor = \frac{T^2}{n} = \frac{1010^2}{12} = 85008.33$

$SST =$ *Total sum of the squares* $= (\Sigma {X_1}^2 + \Sigma X_2^2 + \Sigma X_3^2) - C.F$

$= (24150 + 27350 + 34350) - 85008.33$

$= 85850 - 85008.33$

$= 841.67$

$SSC =$ *Sum of the squares between the samples*

$= (\Sigma {X_1}^2/n_1 + \Sigma X_2^2/n_2 + \Sigma X_3^2/n_3) - C.F$

$= (310^2/4 + 330^2/4 + 370^2/4) - 85008.33$

$= (24025 + 27225 + 34225) - 85008.33$

$= 466.67$

$SSE = SST - SSC = 841.67 - 466.67 = 375$

Degrees of freedom: $df1 = r - 1 = 3 - 1 = 2$ *and* $df2 = n - r = 12 - 3 = 9$

Thus, $MSC = SSC/df1 = 466.67/2 = 233.34$

and $MSE = SSE/df2 = 375/9 = 41.67$

The table value of F for df1 =2 , $df2 = 9$ *and* $\alpha = 0.05$ *level of significance is 4.26.*

Analysis of Variance (ANOVA) Table is shown in Table 3 which gives $F = 5.60$

$F_{calculated} > F_{critical} \implies$ *Reject Null Hypothesis*

Hence, we conclude that the mean mathematics test scores are not the same for the three teaching methods. Therefore, at least one teaching method produces a significantly different average score.

Table 3: Analysis of Variance (ANOVA) Table

Source of Variation	Sum of squares (SS)	Degrees of freedom	Mean Squares (MS)	Variance ratio of F
Between samples	SSC = 466.67	$df1 = 2$	$MSC = \frac{SSC}{df1} = \frac{466.67}{2} = 233.34$	$F = \frac{MSC}{MSE} = \frac{233.34}{41.67} = 5.60$
Within samples	SSE = 375	$df2 = 9$	$MSE = \frac{SSE}{df2} = \frac{375}{9} = 41.67$	
Total	SST = 841.67	n -1 = 11		

8. Practice Problems

1. A random sample of 64 bags of white cheddar popcorn weighed, on average, 5.23 ounces with a standard deviation of 0.24 ounce. Test the hypothesis that $\mu = 5.5$ ounces against the alternative hypothesis, $\mu < 5.5$ ounces, at the 0.05 level of significance. You may assume $t_{0.05,63} = 1.669$.

2. An electrical firm manufactures light bulbs that have a lifetime that is approximately normally distributed with a mean of 800 hours and a standard deviation of 40 hours. Test the hypothesis that $\mu = 800$ hours against the alternative, $\mu \neq 800$, if a random sample of 30 bulbs has an average life of 788hours. Use a P-value in your answer. You may assume $z_{0.025} = 1.96$

3. The average height of females in the freshman class of a certain college has historically been162.5 centimeters with a standard deviation of 6.9 centimeters. Is there reason to believe that there has been a change in the average height if a random sample of 50 females in the present freshman class has an average height of 165.2 centimeters? Use a P-value in your conclusion. Assume the standard deviation remains the same. You may assume $z_{0.025} = 1.96$.

4. According to a dietary study, high sodium intake may be related to ulcers, stomach cancer, and migraine headaches. The human requirement for salt is only 220 milligrams per day, which is surpassed in most single servings of ready-to-eat cereals. If a random sample of 20 similar servings of a certain cereal has a mean sodium content of 244 milligrams and a standard deviation of 24.5 milligrams, does this suggest at the 0.05 level of significance that the average sodium content for a single serving of such cereal is greater than 220 milligrams? Assume the distribution of sodium contents to be normal. You may assume $t_{0.05,19} = 1.729$.

5. An experiment was conducted to compare the alcohol content of soy sauce on two different production lines. Production was monitored eight time s a day. The data are shown here.

   Production line 1: 0.48 0.39 0.42 0.52 0.40 0.48 0.52 0.52

   Production line 2: 0.38 0.37 0.39 0.41 0.38 0.39 0.40 0.39

Assume both populations are normal. It is suspected that production line 1 is not producing as consistently as production line 2 in terms of alcohol content. Test the hypothesis that $\sigma_1 = \sigma_2$ against the alternative that $\sigma_1 \neq \sigma_2$. Use a P-value.

6. Hydrocarbon emissions from cars are known to have decreased dramatically during the 1980s. A study was conducted to compare the hydrocarbon emissions at idling speed, in parts per million (ppm), for automobiles from 1980 and 1990. Twenty cars of each model year were randomly selected, and their hydrocarbon emission levels were recorded. The data are as follows:

   1980 models:

   141 359 247 940 882 494 306 210 105 880 200 223 188 940 241 190 300 435 241 380

   1990 models:

   140 160 20 20 223 60 20 95 360 70 220 400 217 58 235 380 200 175 85 65

   Test the hypothesis that $\sigma_1 = \sigma_2$ against the alternative that $\sigma_1 \neq \sigma_2$. Use a P-value.

7. A clinical psychologist has run a between-subjects experiment comparing two treatments for depression (cognitive-behavioral therapy (CBT) and client-centered therapy (CCT) against a control condition. Subjects were randomly assigned to the experimental condition. After 12 weeks, the subject's depression scores were measured using the CESD depression scale. The data are summarized as follows:

	n	**mean**	**sd**
Control	40	21.4	4.5
CBT	40	16.9	5.5
CCT	40	19.1	5.8

   Use a oneway ANOVA with $\alpha = 0.01$ for the test.

8. A researcher wishes to try three different techniques to lower the blood pressure of individuals diagnosed with high blood pressure. The subjects are randomly assigned to three groups; the first group takes medication, the second group exercises, and the third group follows a special diet. After four weeks, the reduction in each person's blood pressure is recorded. Assume that the data in each group is approximately normal. Is there a significant difference between the techniques used to lower blood pressure, at $\alpha = 0.05$?

Medication group	Exercise group	Diet group
9	9	4
10	2	5
12	3	8
13	6	9
15	8	12

9. A turkey farmer tested three kinds of poultry feeds with the weights (in pounds) of the grown turkeys in each sample given below. Test at $\alpha = 0.05$ whether there is a difference in the mean weights of turkeys consuming the different feeds. If you find a difference, perform pairwise tests to determine where the difference lies. Assume the distributions are normal and variances are equal.

Feed A	Feed B	Feed C
12.3	12.1	11.5
11.4	13.4	12.4
13.4	14.0	10.8
12.5	13.6	12.6
12.0	12.8	11.8
13.1	14.2	11.9

10. A researcher wishes to see whether there is any difference in the weight gains of athletes following one of three special diets. Athletes are randomly assigned to three groups and placed on the diet for 6 weeks. The weight gains (in pounds) are given. Assume weight gains are normally distributed and the variances are equal. At $\alpha = 0.05$ significance level, can the researcher conclude that there is a difference in the diets?

Diet A	Diet B	Diet C
3	10	8
6	12	3
7	11	2
4	14	5
	8	
	6	

F Distribution Table

d.f.N. = degrees of freedom in numerator
d.f.D. = degrees of freedom in denominator

	$\alpha = 0.005$																		
	d.f.N.																		
d.f.D.	***1***	***2***	***3***	***4***	***5***	***6***	***7***	***8***	***9***	***10***	***12***	***15***	***20***	***24***	***30***	***40***	***60***	***120***	**∞**
1	16211	20000	21615	22500	23056	23437	23715	23925	24091	24224	24426	24630	24836	24940	25044	25148	25253	25359	25465
2	198.5	199.0	199.2	199.2	199.3	199.3	199.4	199.4	199.4	199.4	199.4	199.4	199.4	199.5	199.5	199.5	199.5	199.5	199.5
3	55.55	49.80	47.47	46.19	45.39	44.84	44.43	44.13	43.88	43.69	43.39	43.08	42.78	42.62	42.47	42.31	42.15	41.99	41.83
4	31.33	26.28	24.26	23.15	22.46	21.97	21.62	21.35	21.14	20.97	20.70	20.44	20.17	20.03	19.89	19.75	19.61	19.47	19.32
5	22.78	18.31	16.53	15.56	14.94	14.51	14.20	13.96	13.77	13.62	13.38	13.15	12.90	12.78	12.66	12.53	12.40	12.27	12.14
6	18.63	14.54	12.92	12.03	11.46	11.07	10.79	10.57	10.39	10.25	10.03	9.81	9.59	9.47	9.36	9.24	9.12	9.00	8.88
7	16.24	12.40	10.88	10.05	9.52	9.16	8.89	8.68	8.51	8.38	8.18	7.97	7.75	7.65	7.53	7.42	7.31	7.19	7.08
8	14.69	11.04	9.60	8.81	8.30	7.95	7.69	7.50	7.34	7.21	7.01	6.81	6.61	6.50	6.40	6.29	6.18	6.06	5.95
9	13.61	10.11	8.72	7.96	7.47	7.13	6.88	6.69	6.54	6.42	6.23	6.03	5.83	5.73	5.62	5.52	5.41	5.30	5.19
10	12.83	9.43	8.08	7.34	6.87	6.54	6.30	6.12	5.97	5.85	5.66	5.47	5.27	5.17	5.07	4.97	4.86	4.75	4.64
11	12.23	8.91	7.60	6.88	6.42	6.10	5.86	5.68	5.54	5.42	5.24	5.05	4.86	4.76	4.65	4.55	4.44	4.34	4.23
12	11.75	8.51	7.23	6.52	6.07	5.76	5.52	5.35	5.20	5.09	4.91	4.72	4.53	4.43	4.33	4.23	4.12	4.01	3.90
13	11.37	8.19	6.93	6.23	5.79	5.48	5.25	5.08	4.94	4.82	4.64	4.46	4.27	4.17	4.07	3.97	3.87	3.76	3.65
14	11.06	7.92	6.68	6.00	5.56	5.26	5.03	4.86	4.72	4.60	4.43	4.25	4.06	3.96	3.86	3.76	3.66	3.55	3.44
15	10.80	7.70	6.48	5.80	5.37	5.07	4.85	4.67	4.54	4.42	4.25	4.07	3.88	3.79	3.69	3.58	3.48	3.37	3.26
16	10.58	7.51	6.30	5.64	5.21	4.91	4.69	4.52	4.38	4.27	4.10	3.92	3.73	3.64	3.54	3.44	3.33	3.22	3.11
17	10.38	7.35	6.16	5.50	5.07	4.78	4.56	4.39	4.25	4.14	3.97	3.79	3.61	3.51	3.41	3.31	3.21	3.10	2.98
18	10.22	7.21	6.03	5.37	4.96	4.66	4.44	4.28	4.14	4.03	3.86	3.68	3.50	3.40	3.30	3.20	3.10	2.99	2.87
19	10.07	7.09	5.92	5.27	4.85	4.56	4.34	4.18	4.04	3.93	3.76	3.59	3.40	3.31	3.21	3.11	3.00	2.89	2.78
20	9.94	6.99	5.82	5.17	4.76	4.47	4.26	4.09	3.96	3.85	3.68	3.50	3.32	3.22	3.12	3.02	2.92	2.81	2.69
21	9.83	6.89	5.73	5.09	4.68	4.39	4.18	4.01	3.88	3.77	3.60	3.43	3.24	3.15	3.05	2.95	2.84	2.73	2.61
22	9.73	6.81	5.65	5.02	4.61	4.32	4.11	3.94	3.81	3.70	3.54	3.36	3.18	3.08	2.98	2.88	2.77	2.66	2.55
23	9.63	6.73	5.58	4.95	4.54	4.26	4.05	3.88	3.75	3.64	3.47	3.30	3.12	3.02	2.92	2.82	2.71	2.60	2.48
24	9.55	6.66	5.52	4.89	4.49	4.20	3.99	3.83	3.69	3.59	3.42	3.25	3.06	2.97	2.87	2.77	2.66	2.55	2.43
25	9.48	6.60	5.46	4.84	4.43	4.15	3.94	3.78	3.63	3.54	3.37	3.20	3.01	2.92	2.82	2.72	2.61	2.50	2.38
26	9.41	6.54	5.41	4.79	4.38	4.10	3.89	3.73	3.60	3.49	3.33	3.15	2.97	2.87	2.77	2.67	2.56	2.45	2.33
27	9.34	6.49	5.36	4.74	4.34	4.06	3.85	3.69	3.56	3.45	3.28	3.11	2.93	2.83	2.73	2.63	2.52	2.41	2.25
28	9.28	6.44	5.32	4.70	4.30	4.02	3.81	3.65	3.52	3.41	3.25	3.07	2.89	2.79	2.69	2.59	2.48	2.37	2.29
29	9.23	6.40	5.28	4.66	4.26	3.98	3.77	3.61	3.48	3.38	3.21	3.04	2.86	2.76	2.66	2.56	2.45	2.33	2.24
30	9.18	6.35	5.24	4.62	4.23	3.95	3.74	3.58	3.45	3.34	3.18	3.01	2.82	2.73	2.63	2.52	2.42	2.30	2.18
40	8.83	6.07	4.98	4.37	3.99	3.71	3.51	3.35	3.22	3.12	2.95	2.78	2.60	2.50	2.40	2.30	2.18	2.06	1.93
60	8.49	5.79	4.73	4.14	3.76	3.49	3.29	3.13	3.01	2.90	2.74	2.57	2.39	2.29	2.19	2.08	1.96	1.83	1.69
120	8.18	5.54	4.50	3.92	3.55	3.28	3.09	2.93	2.81	2.71	2.54	2.37	2.19	2.09	1.98	1.87	1.75	1.61	1.43
∞	7.88	5.30	4.28	3.72	3.35	3.09	2.90	2.74	2.62	2.52	2.36	2.19	2.00	1.90	1.79	1.67	1.53	1.36	1.00

F Distribution Table

	$\alpha = 0.01$																		
	d.f.N.																		
d.f.D.	***1***	***2***	***3***	***4***	***5***	***6***	***7***	***8***	***9***	***10***	***12***	***15***	***20***	***24***	***30***	***40***	***60***	***120***	**∞**
1	4052	4999.5	5403	5625	5764	5859	5928	5982	6022	6056	6106	6157	6209	6235	6261	6287	6313	6339	6366
2	98.50	99.00	99.17	99.25	99.30	99.33	99.36	99.37	99.39	99.40	99.42	99.43	99.45	99.46	99.47	99.47	99,48	99.49	99.50
3	34.12	30.82	29.46	28.71	28.24	27.91	27.67	27.49	27.35	27.23	27.05	26.87	26.69	26.60	26.50	26.41	26.32	26.22	26.13
4	21.20	18.00	16.69	15.98	15.52	15.21	14.98	14.80	14.66	14.55	14.37	14.20	14.02	13.93	13.84	13.75	13.65	13.56	13.46
5	16.26	13.27	12.06	11.39	10.97	10.67	10.46	10.29	10.16	10.05	9.89	9.72	9.55	9.47	9.38	9.29	9.20	9.11	9.02
6	13.75	10.92	9.78	9.15	8.75	8.47	8.26	8.10	7.98	7.87	7.72	7.56	7.40	7.31	7.23	7.14	7.06	6.97	6.88
7	12.25	9.55	8.45	7.85	7.46	7.19	6.99	6.84	6.72	6.62	6.47	6.31	6.16	6.07	5.99	5.91	5.82	5.74	5.65
8	11.26	8.65	7.59	7.01	6.63	6.37	6.18	6.03	5.91	5.81	5.67	5.52	5.36	5.28	5.20	5.12	5.03	4.95	4.86
9	10.56	8.02	6.99	6.42	6.06	5.80	5.61	5.47	5.35	5.26	5.11	4.96	4.81	4.73	4.65	4.57	4.48	4.40	4.31
10	10.04	7.56	6.55	5.99	5.64	5.39	5.20	5.06	4.94	4.85	4.71	4.56	4.41	4.33	4.25	4.17	4.08	4.00	3.91
11	9.65	7.21	6.22	5.67	5.32	5.07	4.89	4.74	4.63	4.54	4.40	4.25	4.10	4.02	3.94	3.86	3.78	3.69	3.60
12	9.33	6.93	5.95	5.41	5.06	4.82	4.64	4.50	4.39	4.30	4.16	4.01	3.86	3.78	3.70	3.62	3.54	3.45	3.36
13	9.07	6.70	5.74	5.21	4.86	4.62	4.44	4.30	4.19	4.10	3.96	3.82	3.66	3.59	3.51	3.43	3.34	3.25	3.17
14	8.86	6.51	5.56	5.04	4.69	4.46	4.28	4.14	4.03	3.94	3.80	3.66	3.51	3.43	3.35	3.27	3.18	3.09	3.00
15	8.68	6.36	5.42	4.89	4.56	4.32	4.14	4.00	3.89	3.80	3.67	3.52	3.37	3.29	3.21	3.13	3.05	2.96	2.87
16	8.53	6.23	5.29	4.77	4.44	4.20	4.03	3.89	3.78	3.69	3.55	3.41	3.26	3.18	3.10	3.02	2.93	2.84	2.75
17	8.40	6.11	5.18	4.67	4.34	4.10	3.93	3.79	3.68	3.59	3.46	3.31	3.16	3.08	3.00	2.92	2.83	2.75	2.65
18	8.29	6.01	5.09	4.58	4.25	4.01	3.84	3.71	3.60	3.51	3.37	3.23	3.08	3.00	2.92	2.84	2.75	2.66	2.57
19	8.18	5.93	5.01	4.50	4.17	3.94	3.77	3.63	3.52	3.43	3.30	3.15	3.00	2.92	2.84	2.76	2.67	2.58	2.49
20	8.10	5.85	4.94	4.43	4.10	3.87	3.70	3.56	3.46	3.37	3.23	3.09	2.94	2.86	2.78	2.69	2.61	2.52	2.42
21	8.02	5.78	4.87	4.37	4.04	3.81	3.64	3.51	3.40	3.31	3.17	3.03	2.88	2.80	2.72	2.64	2.55	2.46	2.36
22	7.95	5.72	4.82	4.31	3.99	3.76	3.59	3.45	3.35	3.26	3.12	2.98	2.83	2.75	2.67	2.58	2.50	2.40	2.31
23	7.88	5.66	4.76	4.26	3.94	3.71	3.54	3.41	3.30	3.21	3.07	2.93	2.78	2.70	2.62	2.54	2.45	2.35	2.26
24	7.82	5.61	4.72	4.22	3.90	3.67	3.50	3.36	3.26	3.17	3.03	2.89	2.74	2.66	2.58	2.49	2.40	2.31	2.21
25	7.77	5.57	4.68	4.18	3.85	3.63	3.46	3.32	3.22	3.13	2.99	2.85	2.70	2.62	2.54	2.45	2.36	2.27	2.17
26	7.72	5.53	4.64	4.14	3.82	3.59	3.42	3.29	3.18	3.09	2.96	2.81	2.66	2.58	2.50	2.42	2.33	2.23	2.13
27	7.68	5.49	4.60	4.11	3.78	3.56	3.39	3.26	3.15	3.06	2.93	2.78	2.63	2.55	2.47	2.38	2.29	2.20	2.10
28	7.64	5.45	4.57	4.07	3.75	3.53	3.36	3.23	3.12	3.03	2.90	2.75	2.60	2.52	2.44	2.35	2.26	2.17	2.06
29	7.60	5.42	4.54	4.04	3.73	3.50	3.33	3.20	3.09	3.00	2.87	2.73	2.57	2.49	2.41	2.33	2.23	2.14	2.03
30	7.56	5.39	4.51	4.02	3.70	3.47	3.30	3.17	3.07	2.98	2.84	2.70	2.55	2.47	2.39	2.30	2.21	2.11	2.01
40	7.31	5.18	4.31	3.83	3.51	3.29	3.12	2.99	2.89	2.80	2.66	2.52	2.37	2.29	2.20	2.11	2.02	1.92	1.80
60	7.08	4.98	4.13	3.65	3.34	3.12	2.95	2.82	2.72	2.63	2.50	2.35	2.20	2.12	2.03	1.94	1.84	1.73	1.60
120	6.85	4.79	3.95	3.48	3.17	2.96	2.79	2.66	2.56	2.47	2.34	2.19	2.03	1.95	1.86	1.76	1.66	1.53	1.38
∞	6.63	4.61	3.78	3.32	3.02	2.80	2.64	2.51	2.41	2.32	2.18	2.04	1.88	1.79	1.70	1.59	1.47	1.32	1.00

F Distribution Table

	$\alpha = 0.025$																		
	d.f.N.																		
d.f.D.	***1***	***2***	***3***	***4***	***5***	***6***	***7***	***8***	***9***	***10***	***12***	***15***	***20***	***24***	***30***	***40***	***60***	***120***	**∞**
1	647.8	799.5	864.2	899.6	921.8	937.1	948.2	956.7	963.3	968.6	976.7	984.9	993.1	997.2	1001	1006	1010	1014	1018
2	38.51	39.00	39.17	39.25	39.30	39.33	39.36	39.37	39.39	39.40	39.41	39.43	39.45	39.46	39.46	39.47	39.48	39.49	39.50
3	17.44	16.04	15.44	15.10	14.88	14.73	14.62	14.54	14.47	14.42	14.34	14.25	14.17	14.12	14.08	14.04	13.99	13.95	13.90
4	12.22	10.65	9.98	9.60	9.36	9.20	9.07	8.98	8.90	8.84	8.75	8.66	8.56	8.51	8.46	8.41	8.36	8.31	8.26
5	10.01	8.43	7.76	7.39	7.15	6.98	6.85	6.76	6.68	6.62	6.52	6.43	6.33	6.28	6.23	6.18	6.12	6.07	6.02
6	8.81	7.26	6.60	6.23	5.99	5.82	5.70	5.60	5.52	5.46	5.37	5.27	5.17	5.12	5.07	5.01	4.96	4.90	4.85
7	8.07	6.54	5.89	5.52	5.29	5.12	4.99	4.90	4.82	4.76	4.67	4.57	4.47	4.42	4.36	4.31	4.25	4.20	4.14
8	7.57	6.06	5.42	5.05	4.82	4.65	4.53	4.43	4.36	4.30	4.20	4.10	4.00	3.95	3.89	3.84	3.78	3.73	3.67
9	7.21	5.71	5.08	4.72	4.48	4.32	4.20	4.10	4.03	3.96	3.87	3.77	3.67	3.61	3.56	3.51	3.45	3.39	3.33
10	6.94	5.46	4.83	4.47	4.24	4.07	3.95	3.85	3.78	3.72	3.62	3.52	3.42	3.37	3.31	3.26	3.20	3.14	3.08
11	6.72	5.26	4.63	4.28	4.04	3.88	3.76	3.66	3.59	3.53	3.43	3.33	3.23	3.17	3.12	3.06	3.00	2.94	2.88
12	6.55	5.10	4.47	4.12	3.89	3.73	3.61	3.51	3.44	3.37	3.28	3.18	3.07	3.02	2.96	2.91	2.85	2.79	2.72
13	6.41	4.97	4.35	4.00	3.77	3.60	3.48	3.39	3.31	3.25	3.15	3.05	2.95	2.89	2.84	2.78	2.72	2.66	2.60
14	6.30	4.86	4.24	3.89	3.66	3.50	3.38	3.29	3.21	3.15	3.05	2.95	2.84	2.79	2.73	2.67	2.61	2.55	2.49
15	6.20	4.77	4.15	3.80	3.58	3.41	3.29	3.20	3.12	3.06	2.96	2.86	2.76	2.70	2.64	2.59	2.52	2.46	2.40
16	6.12	4.69	4.08	3.73	3.50	3.34	3.22	3.12	3.05	2.99	2.89	2.79	2.68	2.63	2.57	2.51	2.45	2.38	2.32
17	6.04	4.62	4.01	3.66	3.44	3.28	3.16	3.06	2.98	2.92	2.82	2.72	2.62	2.56	2.50	2.44	2.38	2.32	2.25
18	5.98	4.56	3.95	3.61	3.38	3.22	3.10	3.01	2.93	2.87	2.77	2.67	2.56	2.50	2.44	2.38	2.32	2.26	2.19
19	5.92	4.51	3.90	3.56	3.33	3.17	3.05	2.96	2.88	2.82	2.72	2.62	2.51	2.45	2.39	2.33	2.27	2.20	2.13
20	5.87	4.46	3.86	3.51	3.29	3.13	3.01	2.91	2.84	2.77	2.68	2.57	2.46	2.41	2.35	2.29	2.22	2.16	2.09
21	5.83	4.42	3.82	3.48	3.25	3.09	2.97	2.87	2.80	2.73	2.64	2.53	2.42	2.37	2.31	2.25	2.18	2.11	2.04
22	5.79	4.38	3.78	3.44	3.22	3.05	2.93	2.84	2.76	2.70	2.60	2.50	2.39	2.33	2.27	2.21	2.14	2.08	2.00
23	5.75	4.35	3.75	3.41	3.18	3.02	2.90	2.81	2.73	2.67	2.57	2.47	2.36	2.30	2.24	2.18	2.11	2.04	1.97
24	5.72	4.32	3.72	3.38	3.15	2.99	2.87	2.78	2.70	2.64	2.54	2.44	2.33	2.27	2.21	2.15	2.08	2.01	1.94
25	5.69	4.29	3.69	3.35	3.13	2.97	2.85	2.75	2.68	2.61	2.51	2.41	2.30	2.24	2.18	2.12	2.05	1.98	1.91
26	5.66	4.27	3.67	3.33	3.10	2.94	2.82	2.73	2.65	2.59	2.49	2.39	2.28	2.22	2.16	2.09	2.03	1.95	1.88
27	5.63	4.24	3.65	3.31	3.08	2.92	2.80	2.71	2.63	2.57	2.47	2.36	2.25	2.19	2.13	2.07	2.00	1.93	1.85
28	5.61	4.22	3.63	3.29	3.06	2.90	2.78	2.69	2.61	2.55	2.45	2.34	2.23	2.17	2.11	2.05	1.98	1.91	1.83
29	5.59	4.20	3.61	3.27	3.04	2.88	2.76	2.67	2.59	2.53	2.43	2.32	2.21	2.15	2.09	2.03	1.96	1.89	1.81
30	5.57	4.18	3.69	3.25	3.03	2.87	2.75	2.65	2.57	2.51	2.41	2.31	2.20	2.14	2.07	2.01	1.94	1.87	1.79
40	5.42	4.05	3.46	3.13	2.90	2.74	2.62	2.53	2.45	2.39	2.29	2.18	2.07	2.01	1.94	1.88	1.80	1.72	1.64
60	5.29	3.93	3.34	3.01	2.79	2.63	2.51	2.41	2.33	2.27	2.17	2.06	1.94	1.88	1.82	1.74	1.67	1.58	1.48
120	5.15	3.80	3.23	2.89	2.67	2.52	2.39	2.30	2.22	2.16	2.05	1.94	1.82	1.76	1.69	1.61	1.53	1.43	1.31
∞	5.02	3.69	3.12	2.79	2.57	2.41	2.29	2.19	2.11	2.05	1.94	1.83	1.71	1.64	1.57	1.48	1.39	1.27	1.00

F Distribution Table

	$\alpha = 0.05$																		
	d.f.N.																		
d.f.D.	***1***	***2***	***3***	***4***	***5***	***6***	***7***	***8***	***9***	***10***	***12***	***15***	***20***	***24***	***30***	***40***	***60***	***120***	**∞**
1	161.4	199.5	215.7	224.6	230.2	234.0	236.8	238.9	240.5	241.9	243.9	245.9	248.0	249.1	250.1	251.1	252.2	253.3	254.3
2	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38	19.40	19.41	19.43	19.45	19.45	19.46	19.47	19.48	19.49	19.50
3	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79	8.74	8.70	8.66	8.64	8.62	8.59	8.57	8.55	8.53
4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96	5.91	5.86	5.80	5.77	5.75	5.72	5.69	5.66	5.63
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74	4.68	4.62	4.56	4.53	4.50	4.46	4.43	4.40	4.36
6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06	4.00	3.94	3.87	3.84	3.81	3.77	3.74	3.70	3.67
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64	3.57	3.51	3.44	3.41	3.38	3.34	3.30	3.27	3.23
8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35	3.28	3.22	3.15	3.12	3.08	3.04	3.01	2.97	2.93
9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14	3.07	3.01	2.94	2.90	2.86	2.83	2.79	2.75	2.71
10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98	2.91	2.85	2.77	2.74	2.70	2.66	2.62	2.58	2.54
11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85	2.79	2.72	2.65	2.61	2.57	2.53	2.49	2.45	2.40
12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75	2.69	2.62	2.54	2.51	2.47	2.43	2.38	2.34	2.30
13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71	2.67	2.60	2.53	2.46	2.42	2.38	2.34	2.30	2.25	2.21
14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65	2.60	2.53	2.46	2.39	2.35	2.31	2.27	2.22	2.18	2.13
15	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59	2.54	2.48	2.40	2.33	2.29	2.25	2.20	2.16	2.11	2.07
16	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54	2.49	2.42	2.35	2.28	2.24	2.19	2.15	2.11	2.06	2.01
17	4.45	3.59	3.20	2.96	2.81	2.70	2.61	2.55	2.49	2.45	2.38	2.31	2.23	2.19	2.15	2.10	2.06	2.01	1.96
18	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46	2.41	2.34	2.27	2.19	2.15	2.11	2.06	2.02	1.97	1.92
19	4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42	2.38	2.31	2.23	2.16	2.11	2.07	2.03	1.98	1.93	1.88
20	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39	2.35	2.28	2.20	2.12	2.08	2.04	1.99	1.95	1.90	1.84
21	4.32	3.47	3.07	2.84	2.68	2.57	2.49	2.42	2.37	2.32	2.25	2.18	2.10	2.05	2.01	1.96	1.92	1.87	1.81
22	4.30	3.44	3.05	2.82	2.66	2.55	2.46	2.40	2.34	2.30	2.23	2.15	2.07	2.03	1.98	1.94	1.89	1.84	1.78
23	4.28	3.42	3.03	2.80	2.64	2.53	2.44	2.37	2.32	2.27	2.20	2.13	2.05	2.01	1.96	1.91	1.86	1.81	1.76
24	4.26	3.40	3.01	2.78	2.62	2.51	2.42	2.36	2.30	2.25	2.18	2.11	2.03	1.98	1.94	1.89	1.84	1.79	1.73
25	4.24	3.39	2.99	2.76	2.60	2.49	2.40	2.34	2.28	2.24	2.16	2.09	2.01	1.96	1.92	1.87	1.82	1.77	1.71
26	4.23	3.37	2.98	2.74	2.59	2.47	2.39	2.32	2.27	2.22	2.15	2.07	1.99	1.95	1.90	1.85	1.80	1.75	1.69
27	4.21	3.35	2.96	2.73	2.57	2.46	2.37	2.31	2.25	2.20	2.13	2.06	1.97	1.93	1.88	1.84	1.79	1.73	1.67
28	4.20	3.34	2.95	2.71	2.56	2.45	2.36	2.29	2.24	2.19	2.12	2.04	1.96	1.91	1.87	1.82	1.77	1.71	1.65
29	4.18	3.33	2.93	2.70	2.55	2.43	2.35	2.28	2.22	2.18	2.10	2.03	1.94	1.90	1.85	1.81	1.75	1.70	1.64
30	4.17	3.32	2.92	2.69	2.53	2.42	2.33	2.27	2.21	2.16	2.09	2.01	1.93	1.89	1.84	1.79	1.74	1.68	1.62
40	4.08	3.23	2.84	2.61	2.45	2.34	2.25	2.18	2.12	2.08	2.00	1.92	1.84	1.79	1.74	1.69	1.64	1.58	1.51
60	4.00	3.15	2.76	2.52	2.37	2.25	2.17	2.10	2.04	1.99	1.92	1.84	1.75	1.70	1.65	1.59	1.53	1.47	1.39
120	3.92	3.07	2.68	2.45	2.29	2.17	2.09	2.02	1.96	1.91	1.83	1.75	1.66	1.61	1.55	1.50	1.43	1.35	1.25
∞	3.84	3.00	2.60	2.37	2.21	2.10	2.01	1.94	1.88	1.83	1.75	1.67	1.57	1.52	1.46	1.39	1.32	1.22	1.00

STANDARD NORMAL DISTRIBUTION: Table Values Represent AREA to the LEFT of the Z score.

Z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.9	.00005	.00005	.00004	.00004	.00004	.00004	.00004	.00004	.00003	.00003
-3.8	.00007	.00007	.00007	.00006	.00006	.00006	.00006	.00005	.00005	.00005
-3.7	.00011	.00010	.00010	.00010	.00009	.00009	.00008	.00008	.00008	.00008
-3.6	.00016	.00015	.00015	.00014	.00014	.00013	.00013	.00012	.00012	.00011
-3.5	.00023	.00022	.00022	.00021	.00020	.00019	.00019	.00018	.00017	.00017
-3.4	.00034	.00032	.00031	.00030	.00029	.00028	.00027	.00026	.00025	.00024
-3.3	.00048	.00047	.00045	.00043	.00042	.00040	.00039	.00038	.00036	.00035
-3.2	.00069	.00066	.00064	.00062	.00060	.00058	.00056	.00054	.00052	.00050
-3.1	.00097	.00094	.00090	.00087	.00084	.00082	.00079	.00076	.00074	.00071
-3.0	.00135	.00131	.00126	.00122	.00118	.00114	.00111	.00107	.00104	.00100
-2.9	.00187	.00181	.00175	.00169	.00164	.00159	.00154	.00149	.00144	.00139
-2.8	.00256	.00248	.00240	.00233	.00226	.00219	.00212	.00205	.00199	.00193
-2.7	.00347	.00336	.00326	.00317	.00307	.00298	.00289	.00280	.00272	.00264
-2.6	.00466	.00453	.00440	.00427	.00415	.00402	.00391	.00379	.00368	.00357
-2.5	.00621	.00604	.00587	.00570	.00554	.00539	.00523	.00508	.00494	.00480
-2.4	.00820	.00798	.00776	.00755	.00734	.00714	.00695	.00676	.00657	.00639
-2.3	.01072	.01044	.01017	.00990	.00964	.00939	.00914	.00889	.00866	.00842
-2.2	.01390	.01355	.01321	.01287	.01255	.01222	.01191	.01160	.01130	.01101
-2.1	.01786	.01743	.01700	.01659	.01618	.01578	.01539	.01500	.01463	.01426
-2.0	.02275	.02222	.02169	.02118	.02068	.02018	.01970	.01923	.01876	.01831
-1.9	.02872	.02807	.02743	.02680	.02619	.02559	.02500	.02442	.02385	.02330
-1.8	.03593	.03515	.03438	.03362	.03288	.03216	.03144	.03074	.03005	.02938
-1.7	.04457	.04363	.04272	.04182	.04093	.04006	.03920	.03836	.03754	.03673
-1.6	.05480	.05370	.05262	.05155	.05050	.04947	.04846	.04746	.04648	.04551
-1.5	.06681	.06552	.06426	.06301	.06178	.06057	.05938	.05821	.05705	.05592
-1.4	.08076	.07927	.07780	.07636	.07493	.07353	.07215	.07078	.06944	.06811
-1.3	.09680	.09510	.09342	.09176	.09012	.08851	.08691	.08534	.08379	.08226
-1.2	.11507	.11314	.11123	.10935	.10749	.10565	.10383	.10204	.10027	.09853
-1.1	.13567	.13350	.13136	.12924	.12714	.12507	.12302	.12100	.11900	.11702
-1.0	.15866	.15625	.15386	.15151	.14917	.14686	.14457	.14231	.14007	.13786
-0.9	.18406	.18141	.17879	.17619	.17361	.17106	.16853	.16602	.16354	.16109
-0.8	.21186	.20897	.20611	.20327	.20045	.19766	.19489	.19215	.18943	.18673
-0.7	.24196	.23885	.23576	.23270	.22965	.22663	.22363	.22065	.21770	.21476
-0.6	.27425	.27093	.26763	.26435	.26109	.25785	.25463	.25143	.24825	.24510
-0.5	.30854	.30503	.30153	.29806	.29460	.29116	.28774	.28434	.28096	.27760
-0.4	.34458	.34090	.33724	.33360	.32997	.32636	.32276	.31918	.31561	.31207
-0.3	.38209	.37828	.37448	.37070	.36693	.36317	.35942	.35569	.35197	.34827
-0.2	.42074	.41683	.41294	.40905	.40517	.40129	.39743	.39358	.38974	.38591
-0.1	.46017	.45620	.45224	.44828	.44433	.44038	.43644	.43251	.42858	.42465
-0.0	.50000	.49601	.49202	.48803	.48405	.48006	.47608	.47210	.46812	.46414

STANDARD NORMAL DISTRIBUTION: Table Values Represent AREA to the LEFT of the Z score.

Z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.50000	.50399	.50798	.51197	.51595	.51994	.52392	.52790	.53188	.53586
0.1	.53983	.54380	.54776	.55172	.55567	.55962	.56356	.56749	.57142	.57535
0.2	.57926	.58317	.58706	.59095	.59483	.59871	.60257	.60642	.61026	.61409
0.3	.61791	.62172	.62552	.62930	.63307	.63683	.64058	.64431	.64803	.65173
0.4	.65542	.65910	.66276	.66640	.67003	.67364	.67724	.68082	.68439	.68793
0.5	.69146	.69497	.69847	.70194	.70540	.70884	.71226	.71566	.71904	.72240
0.6	.72575	.72907	.73237	.73565	.73891	.74215	.74537	.74857	.75175	.75490
0.7	.75804	.76115	.76424	.76730	.77035	.77337	.77637	.77935	.78230	.78524
0.8	.78814	.79103	.79389	.79673	.79955	.80234	.80511	.80785	.81057	.81327
0.9	.81594	.81859	.82121	.82381	.82639	.82894	.83147	.83398	.83646	.83891
1.0	.84134	.84375	.84614	.84849	.85083	.85314	.85543	.85769	.85993	.86214
1.1	.86433	.86650	.86864	.87076	.87286	.87493	.87698	.87900	.88100	.88298
1.2	.88493	.88686	.88877	.89065	.89251	.89435	.89617	.89796	.89973	.90147
1.3	.90320	.90490	.90658	.90824	.90988	.91149	.91309	.91466	.91621	.91774
1.4	.91924	.92073	.92220	.92364	.92507	.92647	.92785	.92922	.93056	.93189
1.5	.93319	.93448	.93574	.93699	.93822	.93943	.94062	.94179	.94295	.94408
1.6	.94520	.94630	.94738	.94845	.94950	.95053	.95154	.95254	.95352	.95449
1.7	.95543	.95637	.95728	.95818	.95907	.95994	.96080	.96164	.96246	.96327
1.8	.96407	.96485	.96562	.96638	.96712	.96784	.96856	.96926	.96995	.97062
1.9	.97128	.97193	.97257	.97320	.97381	.97441	.97500	.97558	.97615	.97670
2.0	.97725	.97778	.97831	.97882	.97932	.97982	.98030	.98077	.98124	.98169
2.1	.98214	.98257	.98300	.98341	.98382	.98422	.98461	.98500	.98537	.98574
2.2	.98610	.98645	.98679	.98713	.98745	.98778	.98809	.98840	.98870	.98899
2.3	.98928	.98956	.98983	.99010	.99036	.99061	.99086	.99111	.99134	.99158
2.4	.99180	.99202	.99224	.99245	.99266	.99286	.99305	.99324	.99343	.99361
2.5	.99379	.99396	.99413	.99430	.99446	.99461	.99477	.99492	.99506	.99520
2.6	.99534	.99547	.99560	.99573	.99585	.99598	.99609	.99621	.99632	.99643
2.7	.99653	.99664	.99674	.99683	.99693	.99702	.99711	.99720	.99728	.99736
2.8	.99744	.99752	.99760	.99767	.99774	.99781	.99788	.99795	.99801	.99807
2.9	.99813	.99819	.99825	.99831	.99836	.99841	.99846	.99851	.99856	.99861
3.0	.99865	.99869	.99874	.99878	.99882	.99886	.99889	.99893	.99896	.99900
3.1	.99903	.99906	.99910	.99913	.99916	.99918	.99921	.99924	.99926	.99929
3.2	.99931	.99934	.99936	.99938	.99940	.99942	.99944	.99946	.99948	.99950
3.3	.99952	.99953	.99955	.99957	.99958	.99960	.99961	.99962	.99964	.99965
3.4	.99966	.99968	.99969	.99970	.99971	.99972	.99973	.99974	.99975	.99976
3.5	.99977	.99978	.99978	.99979	.99980	.99981	.99981	.99982	.99983	.99983
3.6	.99984	.99985	.99985	.99986	.99986	.99987	.99987	.99988	.99988	.99989
3.7	.99989	.99990	.99990	.99990	.99991	.99991	.99992	.99992	.99992	.99992
3.8	.99993	.99993	.99993	.99994	.99994	.99994	.99994	.99995	.99995	.99995
3.9	.99995	.99995	.99996	.99996	.99996	.99996	.99996	.99996	.99997	.99997